

Linking fish physiological biomarkers to habitat integrity in the Atlantic Forest

Biological Conservation OR Neotropical Ichthyology

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The Atlantic Forest is one of the world's most important biodiversity hotspots. Despite occupying only a small fraction of its original distribution, it continues to support extraordinary levels of species richness and endemism. At the same time, it is one of the most threatened tropical ecosystems on Earth due to habitat loss, fragmentation, urban expansion, and agricultural development. The present study evaluated whether biomarkers of physiological stress in fish could detect environmental disturbances occurring within and around the Guaribas Biological Reserve (REBIO Guaribas), a strictly protected area of Atlantic Forest in northeastern Brazil. Despite sites within the administrative boundaries of the REBIO showing no signs alteration, from both environmental variables and biomarkers, significant differences in biomarker responses were observed among sampling sites, indicating that environmental conditions are not homogeneous throughout the stream studied. Elevated levels of all biomarkers at sites located outside or near the boundaries of the protected area suggest exposure to environmental stressors potentially associated with anthropogenic activities. These findings support the use of biochemical and physiological biomarkers as sensitive tools for detecting sublethal effects of environmental degradation, even in landscapes that contain legally protected areas.

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1 Introduction

The Atlantic Forest is one of the world's most important biodiversity hotspots (Myers et al., 2000). Despite occupying only a small fraction of its original distribution, it continues to support extraordinary levels of species richness and endemism (Marques et al., 2021). At the same time, it is one of the most threatened tropical ecosystems on Earth due to habitat loss, fragmentation, urban expansion, and agricultural development (Marques & Grelle, 2021). Conservation efforts in the Atlantic Forest have traditionally focused on terrestrial organisms such as plants, birds, and mammals. However, freshwater ecosystems are equally dependent on forest integrity and are often overlooked in conservation planning (Abilhoa et al., 2011; Gouveia et al., 2017).

Padial et al. (2021) mention a wide range of threats to aquatic systems that can affect the Atlantic Forest, ranging from dam construction and flow alterations to the emerging threat of microplastic contamination (Eerkes-Medrano & Thompson, 2018). Pollution, from pesticides, fertilizers, heavy metals, pharmaceuticals or microplastics (Evans, 1987; Harmon, 2018), come mostly from urban and agricultural sources, and untreated urban sewage is amongst the most common (Costa et al., 2020). In agricultural areas, fish farming, agricultural runoff, and the destruction of riparian forests, remain as important threats to aquatic ecosystem integrity (Padial et al., 2021).

Ecological responses to pollution, often become visible only after environmental impacts have already affected freshwater organisms (Macêdo et al., 2019). Furthermore, the toxicity responsible for behavioral change depends on the organism, time of exposure and particle type, which demonstrates the need for physiological studies, specially in natural environments (Amoatey & Baawain, 2019). For these reasons, physiological biomarkers can provide particularly valuable early-warning indication (Ondei et al., 2020), as physiological, morphological and behavioural changes as species are affected in the ecosystem.

Fish are among the most widely used organisms to evaluate the health of aquatic systems, since their biochemical and physiological changes serve as biomarkers of environmental pollution (Cazenave et al., 2009; Ondei et al., 2020). One of the main physiological functions affected by pollutants is the osmoregulation, and its associated pH control. Both are predominantly done by gills, and are critical to gas exchange, ionic regulation and nitrogen excretion (Evans, 1987). The effects of this action on fish can be measured through a few blood markers, mainly lower levels of Chloride and Sodium ions, obtained by freshwater fish through exchange with internal ions (Evans, 1987), and higher levels of plasmatic proteins (Sabae & Mohamed, 2015). This osmoregulation impairment and consequent decrease in plasma ionic concentration can represent an osmotic challenge for tissues. The hydration level of gills and muscles is another physiological parameter associated with an animal's capacity to cope with environmental pollution and is therefore subject to measurable changes induced by toxic compounds present in the environment. This active control was suggested by Ayrapetyan (2012) as a universal biomarker for detecting pollution in the environment (David et al., 2018; Sabae & Mohamed, 2015).

Another relevant biomarker is the abnormal activity of the multixenobiotic resistance (MXR) phenotype (Bard, 2000; Žaja et al., 2006), an innate gene of many aquatic species that is expressed in organs such as gills, kidneys, blood-brain barrier, liver and pancreas. It is activated by exposure to pollutants and results in the expression of P-glycoproteins (Pgp) in the membrane, which act to remove endogenous or exogenous toxic compounds out of the cell, protecting DNA and inducing toxic compounds excretion (Smital & Kurelec, 1998). Although the MXR activity explains how some species are more resistant to the presence of pollutants, some substances can reverse its effectiveness, being called chemosensitizers (or simply MXR-inhibitors) (Smital et al., 2004). Those compounds can interact with the Pgp activity, altering the mechanism regulation, and saturate or even reverse MXR activity. By saturating the Pgp pump, these chemosensitizers allow other xenobiotics to enter the cell and increase its toxicity, even if the chemosensitizers themselves are not toxic. Precisely because they are mostly organic, natural, primarily harmless and very present in the materials emitted anthropogenic in nature, its detection should also be a priority in environmental risk studies (Kurelec et al., 2000).

In the present study, the hypothesis of markers linked to physiological stress of fish providing evidence of the environmental stress was tested along a headwater stream that originates within a protected area - Biological Reserve, or REBIO, according to BRASIL (2000) - and its course extends beyond its administrative boundaries (MMA, 2003). Given the importance of the Atlantic Forest to the biodiversity (Pereira et al., 2026), we aim with this study, (1) to identify physiological stress biomarkers in fish from a headwater stream, (2) to associate these biomarkers to the ecological health of sites inside and surrounding a protected area of Atlantic Forest. This will enable decision makers to take action on the influence of anthropogenic threats and the ecological consequence of pollution exposure of the buffer zone.

2 Material and Methods

Study area and sampling design. This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. The reserve is located in the municipality of Mamanguape, state of Paraíba, in a heterogeneous landscape influenced by the adjacent Caatinga ecoregion. It was created in 1990 to preserve one of the last remnants of the Atlantic Forest in the region (MMA, 2003). Despite a REBIO being one of the most restricted types of protected areas (BRASIL, 2000), there are a number of human settlements in its buffer zone. The management plan for the REBIO states several rural communities directly surrounding the reserve (Caiana, Pepina, Imbiribeira, Brejinho, Água Fria, João Pereira and Piabuçu), in addition to indigenous human populations (mostly the Potiguaras) (MMA, 2003). These populations are important because of their historic patterns of occupation and land use, but also because their agricultural activities, fishing, modification of the natural vegetation for livestock and overall use of the natural resources from the buffer zone directly affect the conservation of the REBIO. This has been aggravated in recent years by large scale use of water and land for cultivation of monocultures, like

sugarcane (Heinrichs et al., 2017). The surrounding landscape is predominantly agricultural, including coconut plantations, livestock farming, orchards subsistence agriculture. These land uses can act as sources of diffuse pollution, especially through pesticide runoff and domestic effluents, creating a spatial gradient of environmental disturbance (MMA, 2003; Soler, 2004). Thus, the landscape is mainly rural and the land use causes direct impact to aquatic systems, by causing siltation of streams and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil (Arruda et al., 2013; MMA, 2026)

The climate in the region is classified as As', characterized by a rainy season from February to July and a dry period from October to December (Peel et al., 2007). The average annual temperature varies from 24 to 26 °C, while the annual rainfall varies between 1,750 and 2,000 mm. The hydrographic network is composed of small headwater streams fed by rain, notably the Barro Branco and Caiana streams, both tributaries of the Camaratuba River (Gouveia et al., 2017).

Sampling was conducted along the Barro Branco stream between November 2023 and February 2024, comprising six sampling sites. The sampling sites were distributed longitudinally to capture environmental variability and gradients of human influence. Sites P1 to P3 were located within the reserve, where more preserved conditions were expected, while sites P4 to P6 were situated in the buffer zone adjacent to areas with greater human influence (Figure 1).

For the sake of comparison, we classified all sites into four categories defined *a priori* (Figure 2). Reference site or Type I site, which are the sites that occur within the protected area and are well preserved, with intact riparian forest and no crop fields nearby (at least 1 km distant), free flowing water (0.25 m/s), high dissolved oxygen concentration (6 mg/L) and temperature below 28 °C. Type II sites, are those that occur within the protected area, but are not well preserved, failing in one or more of the water quality criteria and/or having its riparian vegetation removed or managed. Type III sites are those outside the protected area but meet the water quality criteria for a type I site, and type IV sites are those that fall outside the protected area and are also disturbed, not meeting the water quality criteria for a type I site. Sites type III and IV were intentionally chosen not to meet the riparian forest and crop proximity criteria. Since they fall outside of the REBIO they are not expected to have intact riparian forest or to be at least 1 km distant from a crop field. These categories were latter confirmed using environmental data.

The ecophysiological evaluation of human impact on the forest, was mainly guided by the fish species inventory for the REBIO made by Gouveia et al. (2017). This study lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012). The predominance of small fish inhabiting small basins is thought to lead to low dispersion and consequent isolation of populations (Gouveia et al., 2017), associated with the fact that most species in the study area are phylogenetic close (with the predominance of Characidae) and are widely distributed, offers an appropriate design

to access the entire local ichthyofauna using a few species, namely the genuses *Hemigrammus* and *Astyanax*.

Environmental data collection. In order to quantitatively confirm the classification of site category types (I, II, III and IV), environmental characteristics of each site were assessed based on four sets of variables: (a) site morphology, (b) water quality, (c) sediment composition, and (d) marginal habitat structure. Site morphology was characterized by measuring stream width (cm) and depth (cm) from three randomly selected transects. Catchment-scale variables, such as elevation and stream length, were obtained using a handheld GPS and satellite imagery. Water quality was evaluated through physical and chemical variables measured with portable equipment, including temperature (°C) and dissolved oxygen concentration (mg/L). Water turbidity was determined in the laboratory using a turbidimeter, based on water samples collected at each sampling site. Water velocity (m/s) was estimated using the float method by Maitland (1990). Sediment composition and habitat physical structure were assessed following protocols adapted by Medeiros et al. (2008) from Pusey et al. (2004) and Mugodo et al. (2006). These variables were estimated from 9 to 12 one-meter quadrats distributed along the stream margins (terrestrial-aquatic interface). Within each quadrat, the percentage cover of sediment types (mud, sand, cobbles, small gravel, large gravel, rocks, and bedrock) and littoral and subaquatic habitat structures was visually estimated. Habitat structure variables included macrophytes, marginal grasses, leaf litter, attached algae, filamentous algae, overhanging vegetation, submerged vegetation, small woody debris, large woody debris, and root masses.

Three 250 mL replicate samples of water from each site were taken and refrigerated for concentration analysis (g L^{-1}) of ammonia, nitrite, nitrate, orthophosphate (soluble reactive phosphorus) and total phosphorus. Nutrient concentrations were obtained based on Standard Methods for the Examination of Water (1998) and Mackereth et al. (1978).

Fish collection and maintenance. Fish were collected between October and November 2023 along the Barro Branco Stream at six previously established sampling points, three located inside and three outside the Guaribas Biological Reserve, following a methodology adapted from Gouveia et al. (2017). Sampling was carried out during the day using a beach seine net (4 m long, 1.5 m high, and 5 mm mesh) and a hand net (50 cm wide and 5 mm mesh) operated with sweeping movements in shallow waters, according to Medeiros et al. (2010). Sampling effort was standardized across all collection events and sampling points. Fish sampling was performed randomly, therefore the occurrence of the species at each sampling site and the sample size varied according to natural dynamic of fish populations.

Fish were transferred to plastic bags containing water from the collection site and atmospheric oxygen, and transported to the Laboratory of Animal Ecophysiology (LEFA) at the State University of Paraíba. Fish were allowed to acclimated for a few hours to minimise stress from handling and transportation, and the ecophysiological experiments occurred on the same day of catch (Macêdo et al., 2019).

Experimental design. For analyses related to osmoregulatory function, blood samples were obtained from 78 individuals, including 38 fish collected within the reserve and 40 in external areas. The fish were anesthetized with eugenol, and blood aliquots were collected by cardiac puncture using pipettes with heparinized tips. The samples were kept on ice until laboratory processing and subsequently centrifuged for plasma separation. Due to the reduced volume of plasma obtained as a consequence of the small body size of the specimens, chloride ion analysis could not be performed. Branchial and muscle tissues from the same individuals were collected to determine tissue moisture content. Plasma and tissue samples were stored at -20°C until laboratory analysis was performed. The remaining 111 individuals, comprising 54 specimens from locations within the reserve and 57 from external areas, were used for evaluation of the multixenobiotic resistance (MXR) phenotype, following the methodologies described by David et al. (2018); Macêdo et al. (2019); Santos et al. (2017).

Plasmatic protein dosage. For the the determination of plasmatic protein dosage, the plasma samples were defrosted and an aliquot of 2 μL of each was diluted with 8 μL of distilled water (proportion) to perform the total protein dosage, where the dye was also diluted in proportion (BioRad® protein assay) (Bradford, 1976). The intensity of each sample coloration was measured with a microplate reader (SpectraMax i3) at 595 nm wave length, and compared to a standard curve (BSA protein). The results were transferred to a spreadsheet, where the average value in milligrams of protein dosage was calculated for each sampling site.

Moisture content. For the analysis of moisture content, parameter related to the osmoregulatory function, fish tissue samples (gill and muscle) were individually stored in 2 mL eppendorf tubes (previously weighted), initially weighted with the tissue moist, and, after 24 hours inside a 60°C oven, weighted with the dry tissue and the percentual content of tissue water was calculated (David et al., 2018).

MXR phenotype activity. The multixenobiotic resistance activity was analysed through the rhodamine B accumulation assay adapted from Smital & Kurelec (1998). Rhodamine B is a substrate of P-glycoprotein, the molecular basis of the MXR phenotype. Thus, after exposing the fish to this substrate, the analysis of the amount of rhodamine accumulated in the animal's tissues (mainly the gills) reflects the activity of the MXR phenotype: the greater the accumulation, the lower the activity, and the lower the accumulation, the greater the activity. Therefore, specimens of fish from each sampling site (three inside the Conservative Unity and three outside, total of six sampling sites) were transferred to plastic aquariums containing dechlorinated water and rhodamine B at 2,5 μM concentration, staying in this condition for one hour. The aquarium remained under constant aeration and protected from direct light. After exposition, the fish were anesthetized with eugenol and sacrificed for gill sampling. The tissue samples were allocated in Eppendorfs tubes and weighted in an analytical scale. The weight of moist tissue was obtained by subtracting the weight of the previously weighted empty eppendorf tubes. The tissues were then homogenized with 500 μL of distilled water and the supernatant was transferred in triplicate of 100 μL to a 96-well microplate. The fluorescence intensity of the supernatant (corresponding to the intracellular rhodamine B fluorescence in the tissue) was measured in the microplate reader (SectraMax i3) at 544 nm excitation and

590 nm emission. The fluorescence value was then normalized by the moist tissue weight in milligrammes (Macêdo et al., 2019).

Statistical analysis. Environmental variables were assessed for multivariate collinearity, excluding redundant variables and combining those that conveyed similar information to reduce data redundancy, following the approach described by Sokal & Rohlf (1995).

```
1  #MXR23 - ORGANIZANDO DADOS-----
2  #####....-----
3
4  dev.off() #apaga os graficos, se houver algum
5  rm(list=ls(all=TRUE)) #limpa a memória
6  cat("\014") #limpa o console
7  getwd()
8  library(openxlsx)
9
10 ##CARREGANDO MATRIZES BRUTAS-----
11
12 habitat <- read.xlsx("D:/mxr23_Q/data/rebio23-habitat.xlsx",
13                     rowNames = T,
14                     colNames = T,
15                     sheet = "ambiente",
16                     rows = 2:20)
17
18 habitat[is.na(habitat)] <- 0
19 habitat
20
21 ### REMOVENDO COLUNAS ZERADAS DE HABITAT-----
22 sum <- colSums(habitat)
23 sum
24 zero_sum <- names(which(colSums(habitat) == 0))
25 zero_sum #nomes das colunas zeradas
26 m_part_cols <- habitat[(colSums(habitat) != 0)] #em != a exclamação inverte o sentido
27 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
28 zero_sum2 #nomes das colunas zeradas
29 sum<-colSums(m_part_cols)
30 sum
31
32 t_grps <- read.xlsx("D:/mxr23_Q/data/rebio23-habitat.xlsx",
33                    rowNames = T,
34                    colNames = T,
35                    sheet = "grupos",
36                    rows = 2:20)
```

```

37 t_grps
38
39
40 ##SALVANDO MATRIZES FINAIS----
41
42 write.table(m_part_cols, "m_hab.csv",
43             sep = ";", dec = ".", #"\t",
44             row.names = TRUE,
45             quote = TRUE,
46             append = FALSE)
47 write.table(t_grps, "t_grps.csv",
48             sep = ";", dec = ".", #"\t",
49             row.names = TRUE,
50             quote = TRUE,
51             append = FALSE)
52 t_grps <- read.csv("t_grps.csv",
53                   sep = ";", dec = ".",
54                   row.names = 1,
55                   header = TRUE,
56                   na.strings = NA)
57 m_hab <- read.csv("m_hab.csv",
58                  sep = ";", dec = ".",
59                  row.names = 1,
60                  header = TRUE,
61                  na.strings = NA)
62
63
64 1 #MXR23----
65 2 ####----
66 3
67 4 ##ORGANIZANDO DADOS----
68 5
69 6 dev.off()
70 7 rm(list=ls(all=TRUE))
71 8 cat("\014")
72 9
73 10 t_grps <- read.csv("t_grps.csv",
74                    sep = ";", dec = ".",
75                    row.names = 1,
76                    header = TRUE,
77                    na.strings = NA)
78 11
79 12 m_hab <- read.csv("m_hab.csv",
80                  sep = ";", dec = ".",
81                  row.names = 1,

```

```

18         header = TRUE,
19         na.strings = NA)
20
21 ##CORRELOGRAMA E REMOÇÃO DE VARIÁVEIS REDUNDANTES OU DESNECESSÁRIAS----
22
23 library(psych)
24 colnames(m_hab)
25
26 png("fig-h.hab_pairs.png")
27 pairs.panels(m_hab[,15:24],
28             method = "pearson", # correlation method
29             scale = FALSE, lm = FALSE,
30             hist.col = "#00AFBB", pch = 19,
31             density = TRUE, # show density plots
32             ellipses = TRUE, # show correlation ellipses
33             alpha = 0.5)
34 dev.off()
35
36 cor <- cor(m_hab)
37 cor
38
39 library(corrplot)
40 png("fig-hab_corrplot.png")
41 corrplot(cor, method = "circle")
42 dev.off()
43
44 ##DELETANDO COLINEARES----
45
46 sink(file = "colineares.txt", append = F, split = T)
47 colnames(m_hab)
48 del_cols <- c("m.Depth_max_cm", "m.Depth_mar_cm")
49 m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]
50
51 ##SOMANDO REDUNDANTES----
52
53 m_hab_part$h.algae <- m_hab_part$h.Algae_f + m_hab_part$h.Algae_a
54 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
55                             %in% c("h.Algae_f", "h.Algae_a"))]
56 m_hab_part$h.debris <- m_hab_part$h.Debris_l + m_hab_part$h.Debris_s
57 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
58                             %in% c("h.Debris_l", "h.Debris_s"))]
59
60 colnames(m_hab_part)

```

```

61 m_hab_part
62
63
64 write.table(m_hab_part, "m_hab_part.csv",
65             sep = ";", dec = ".", #"\t",
66             row.names = TRUE,
67             quote = TRUE,
68             append = FALSE)
69 m_hab_part <- read.csv("m_hab_part.csv",
70                       sep = ";", dec = ".",
71                       row.names = 1,
72                       header = TRUE,
73                       na.strings = NA)

```

These variables were subsequently subjected to Principal Component Analysis (PCA) to evaluate multivariate correlations among sites. Prior to analysis, site morphology and water quality variables were square root transformed, whereas sediment composition and the marginal habitat structure (which were measured as percentages) were arcsine square-root transformed after relativization by column total (McCune & Grace, 2002). PCA was performed using the FactoMineR package in R (Lê et al., 2008). All variables were centered and scaled to unit of variance.

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_hab_part <- read.csv("m_hab_part.csv",
16                       sep = ";", dec = ".",
17                       row.names = 1,
18                       header = TRUE,
19                       na.strings = NA)
20
21 colnames(m_hab_part)

```

```

22
23 ###RELATIVIZAÇÕES E TRANSFORMAÇÕES----
24 library(vegan)
25 library(tidyverse)
26
27 # Selecionando variáveis que começam com "w" ou "m"
28 m_q <- m_hab_part[, grep("^w|m", colnames(m_hab_part))]
29
30 # Selecionando variáveis que começam com "s" ou "h"
31 m_h <- m_hab_part[, grep("^s|h", colnames(m_hab_part))]
32
33 # Removendo os dois primeiros caracteres dos nomes das colunas
34 colnames(m_q) <- sub("^..", "", colnames(m_q))
35 colnames(m_h) <- sub("^..", "", colnames(m_h))
36
37 # Transformações
38 m_q <- sqrt(m_q)
39 m_h <- asin(sqrt(decostand(m_h, method = "total", MARGIN = 2)))
40
41 # Juntando tudo em uma única tabela
42 hab_trns <- cbind(m_h, m_q)
43
44 # Salvando m_hab_trns
45 write.table(hab_trns, "m_hab_trns.csv",
46             sep = ";", dec = ".", #"\t",
47             row.names = TRUE,
48             quote = TRUE,
49             append = FALSE)
50 m_hab_trns <- read.table("m_hab_trns.csv",
51                          sep = ";", dec = ".",
52                          row.names = 1,
53                          header = TRUE,
54                          na.strings = NA)

```

All biomarker results were compared with those from Site 1, which served as the reference site for this study. Site 1 was located deep within the protected area, distant from anthropogenic influences, and met the criteria for a Type I site. After ascertaining the non-normality of the study biomarkers, comparisons were performed using one-sided Wilcoxon rank-sum tests to evaluate the hypothesis that biomarker values at each site were significantly greater than the mean observed at the reference site (Zar, 2010). Different fish species were pooled withing sites. All statistical analyses were conducted in the R statistical environment using RStudio (R Core Team, 2017; R Studio Team, 2022).

3 Results

Environmental variables and nutrient concentration. Water velocity ranged from 0 to 0.293 m/s, with the highest values recorded at site 5. Width varied from 1.2 to 14.6 m, while average depth ranged from 13.7 to 44.2 cm. Dissolved oxygen ranged from 5.4 to 8.1 mg/L, and temperature from 25.2 to 27.8 °C. Conductivity varied between 129.0 and 163.3 S/cm, whereas pH ranged from 5.5 to 7.0. Turbidity was generally low, although higher values were observed at sites 5 and 6, reaching 17.5 and 29.6 NTU, respectively. Elevation ranged from 41.5 to 112.8 m a.s.l., while slope varied between 1.0 and 3.0. Mud and sand were the main substrate components across all sampled sites, with average cover ranging from 30 to 100% and from 0 to 70%, respectively. Habitat structure varied along the stream course. The reference site (site 1) was characterized by high contributions of overhanging vegetation (90%) and exclusively muddy substrates. Site 2 exhibited similar characteristics, whereas site 3 showed reduced habitat complexity. Sites 4–6 displayed greater variation in habitat composition. The habitat structures contributing most overall were overhanging vegetation, debris, and root masses, although these contributions varied among sites. Aquatic macrophytes and littoral grass were particularly abundant at site 4, contributing on average 35.8% and 39.2%, respectively, whereas root masses represented the dominant habitat structure at site 5 (76.7%). Sand cover increased downstream, reaching 70% at site 6, where overhanging vegetation was markedly reduced (Table 1).

```
1 #TABELA DE HABITAT----
2
3 m_hab_part <- read.csv("m_hab_part.csv",
4                       sep = ";", dec = ".",
5                       row.names = 1,
6                       header = TRUE,
7                       na.strings = NA)
8 t_grps <- read.csv("t_grps.csv",
9                  sep = ";", dec = ".",
10                 row.names = 1,
11                 header = TRUE,
12                 na.strings = NA)
13
14 library(dplyr)
15 library(tidyr)
16 m_trab <- m_hab_part %>%
17   rename_with(~ gsub("_", ".", .))
18 m <- m_trab %>%
19   group_by(PontoN = t_grps$PontoN) %>%
20   summarise(across(where(is.numeric),
21                     list(mean = mean, min = min, max = max)),
22             .groups = 'drop') %>%
```

```

23   pivot_longer(
24     cols = -c(PontoN),
25     names_to = c("Variable", ".value"),
26     names_sep = "_"
27   )
28
29   m <- as.data.frame(m)
30   m_wide <- m %>%
31     mutate(stat_string = ifelse(Variable == c("m.Vel.m.s"),
32                                   paste0(round(mean, 3), " (", round(min, 3), "-", round(max, 3),
33                                   paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1),
34     unite("Location", PontoN, sep = "_") %>%
35     select(Variable, Location, stat_string) %>%
36     pivot_wider(names_from = Location, values_from = stat_string)
37
38   m_wide
39   m_wide <- as.data.frame(m_wide)
40   m_wide <- m_wide[, c("Variable", "Ponto5", "Ponto6", "Ponto7", "Ponto8", "Ponto9", "Ponto10")]
41   m_wide
42
43   # Salvado m_wide
44   write.table(m_wide, "m_wide_hab.txt",
45     sep = ";", dec = ".", #"\t",
46     row.names = TRUE,
47     quote = TRUE,
48     append = FALSE)
49   m_wide_hab <- read.table("m_wide_hab.txt",
50     sep = ";", dec = ".",
51     row.names = 1,
52     header = TRUE,
53     na.strings = NA)
54
55   # Exportando dados para Excel
56   library(openxlsx)
57   #write.xlsx(m_wide, file = "tabela de habitat.xlsx", rowNames = FALSE)
58   wb <- loadWorkbook("tabela de habitat.xlsx")
59   writeData(wb, sheet = "Sheet 1", x = m_wide)
60   saveWorkbook(wb, "tabela de habitat.xlsx", overwrite = TRUE)
61
62   # Tabela final ajustada no Excel
63   library(readxl)
64   hab_gtable <- read_excel(
65     path = "tabela de habitat.xlsx",

```

```

66     sheet = "tabela_final",
67     range = "A1:G25") #ajustar para o tamanho da tabela final
68 hab_gtttable
69 library(gt)
70 hab_gtttable <- gt(hab_gtttable)
71 hab_gtttable <- sub_missing(hab_gtttable,
72                             columns = everything(), missing_text = "")
73 #hab_gtttable <- fmt_number(hab_gtttable, columns = "F.O.(%)", decimals = 1)
74 hab_gtttable
75 gtsave(hab_gtttable, "gt-hab_gtttable_xlsx.html")
76 saveRDS(hab_gtttable, "gt-hab_gtttable_xlsx.rds")

1  # Escolher sumário de uma variavel
2  m
3  var <- "w.Temp.C"
4  m[m$Variable == var, "mean"] #cada valor de var
5  summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
6
7  # Escolher sumário de um grupo de variáveis do df m
8  vars <- unique(grep("^h\\.", m$Variable, value = TRUE))
9  summaries <- list() #cria uma lista vazia para guardar os sumários
10 # Loop para cada variável do grupo e guarda em summaries
11 for (var in vars) {
12   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
13 }
14 #var is a temporary variable used in the for loop to iterate through
15 #each variable name that starts with "h."
16
17 summaries
18 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
19 round(summary_table, 2)
20 #sink(file = "summary_h.txt", split = TRUE)
21 round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
22 #sink()
23
24 # Tabela limpa
25 #summary_table <- cbind(Variable = rownames(summary_table), summary_table)
26 #rownames(summary_table) <- NULL
27 #colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
28 #summary_table

```

Principal Component Analysis described the overall structure of the study sites and the most important features in separating them in terms of their physical and chemical variables, site

morphometry, sediment composition, and marginal habitat structure Figure 3. PCA explained 52.4% of the variance in the environmental variables, with the first axis (30.9%) showing a clear spatial gradient. This primary axis separated sites P08 and P10—characterized by greater stream width, depth, and sandy substrate—from sites P05, P06, and P07, which were associated with muddy substrates, higher DO, and dense overhanging vegetation. The second axis (21.5%) was primarily driven by local habitat features, separating site P09, which was strongly associated with root masses and water velocity, from sites characterized by attached algae and macrophytes. The overall multivariate conditions did not differ significantly between the interior and the surroundings of the reserve, although specific local variables such as water turbidity, leaf litter cover, and attached algae were important in explaining specific sites across the conservation unit's boundary.

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("t_grps.csv",
11                    sep = ";", dec = ".",
12                    row.names = 1,
13                    header = TRUE,
14                    na.strings = NA)
15 m_hab_part <- read.csv("m_hab_part.csv",
16                       sep = ";", dec = ".",
17                       row.names = 1,
18                       header = TRUE,
19                       na.strings = NA)
20 m_hab_trns <- read.table("m_hab_trns.csv",
21                         sep = ";", dec = ".",
22                         row.names = 1,
23                         header = TRUE,
24                         na.strings = NA)
25
26 #PCA----
27
28 library("FactoMineR")
29 library("factoextra")
30
31 pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)

```

```

32 print(pca)
33
34 eig.val <- get_eigenvalue(pca)
35 eig.val
36
37 sink(file = "pca_cor_matrix.txt", append = FALSE)
38 pca$var$cor
39 sink()
40
41 fviz_eig(pca, addlabels = TRUE) #, ylim = c(0,35)
42
43 var <- get_pca_var(pca)
44 var
45
46 fviz_pca_var(pca, col.var = "black")
47
48 library("corrplot")
49 corrplot(var$cos2, is.corr=FALSE)
50
51 fviz_cos2(pca, choice = "var", axes = 1:2)
52
53 fviz_pca_var(pca, col.var = "cos2",
54             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
55             repel = TRUE # Avoid text overlapping
56 )
57
58 fviz_pca_var(pca, alpha.var = "cos2")
59
60 corrplot(var$contrib, is.corr=FALSE)
61
62 fviz_contrib(pca, choice = "var", axes = 1, top = 10)
63 fviz_contrib(pca, choice = "var", axes = 2, top = 10)
64 fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)
65
66 fviz_pca_var(pca, col.var = "contrib",
67             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
68 )
69 fviz_pca_var(pca, alpha.var = "contrib")
70
71 ##GROUPING BY KMEANS----
72
73 set.seed(123)
74 res.km <- kmeans(var$coord, centers = 3, nstart = 25)

```

```

75 grp <- as.factor(res.km$cluster)
76 # Color variables by groups
77 fviz_pca_var(pca, col.var = grp,
78               palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
79               legend.title = "Cluster")
80
81 ind <- get_pca_ind(pca)
82 ind
83 ind$contrib
84
85 ##BIPLOTS-----
86
87 fviz_pca_ind(pca)
88
89 fviz_pca_ind(pca, col.ind = "cos2",
90               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
91               repel = TRUE #avoid text overlapping (slow if many points)
92 )
93
94 fviz_pca_ind(pca, pointsize = "cos2",
95               pointshape = 21, fill = "#E7B800",
96               repel = TRUE #avoid text overlapping (slow if many points)
97 )
98
99 fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
100              gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
101              repel = TRUE #avoid text overlapping (slow if many points)
102 )
103
104 fviz_cos2(pca, choice = "ind")
105
106 fviz_contrib(pca, choice = "ind", axes = 1:2)
107
108 # Create a random continuous variable of length 23,
109 # Same length as the number of active individuals in the PCA
110 nrow(m_hab_part)
111 set.seed(123)
112 my.cont.var <- t_grps$Ref_site
113 my.cont.var
114 # Color individuals by the continuous variable
115 fviz_pca_ind(pca, col.ind = my.cont.var,
116               gradient.cols = c("blue", "yellow", "red"),
117               legend.title = "Cont.Var",

```

```

118         repel = FALSE)
119
120 fviz_pca_ind(pca,
121             geom.ind = "point", #show points only (nbut not "text")
122             col.ind = as.factor(my.cont.var), #color by groups
123             palette = "grey",
124             addEllipses = TRUE, #concentration ellipses
125             legend.title = "Groups"
126 )
127
128 length(unique(my.cont.var))
129 fviz_pca_ind(pca,
130             geom.ind = "point",
131             col.ind = as.factor(t_grps$Ref_site),
132             palette = "grey",
133             addEllipses = TRUE, ellipse.type = "confidence",
134             mean.point = TRUE,
135             legend.title = "Groups")
136
137 fviz_pca_biplot(pca, repel = TRUE,
138                 col.var = "#2E9FDF", #variables color
139                 col.ind = "#696969" #individuals color
140 )
141
142 fviz_pca_biplot(pca,
143                 col.ind = as.factor(t_grps$Ref_site), palette = "jco",
144                 addEllipses = TRUE, ellipse.type = "euclid",
145                 label = "var",
146                 col.var = "black", repel = TRUE,
147                 legend.title = "Species")
148
149 #GRAFICO FINAL----
150
151 fviz_pca_biplot(pca, repel = TRUE,
152                 col.var = "red", #variables color
153                 col.ind = "black" #individuals color
154 )
155
156 library(factoextra)
157 library(ggplot2)
158
159 # Convert t_grps$UA to a factor to ensure distinct shapes
160 t_grps$Ref_site <- as.factor(t_grps$Ref_site)

```

```

161
162 # Generate the PCA biplot with individuals in black
163 fviz_pca_biplot(pca, repel = TRUE,
164                 col.var = "red", #variables color
165                 col.ind = "black", #all individuals in black
166                 geom = c("point", "text"))
167
168 # Generate the PCA biplot with individuals in black
169 library(ggrepel) #geom_text_repel
170 png("fig-hab_PCA_fviz.png", width = 11, height = 9, units = "in", res = 400)
171 fviz_pca_biplot(pca_filt, repel = TRUE, repel.var = TRUE,
172                 col.var = "red", #variables color
173                 col.ind = "black", #all individuals in black
174                 geom = c("text")) + #c("point", "text", "")remove default indiv
175   geom_point(aes(shape = t_grps$Ref_site), size = 3) + #map shapes to t_grps$UA
176 #   geom_text_repel(aes(label = t_grps$Ref_site), direction = "both") + # direction = "x" or
177   scale_shape_manual(values = c(15, 16, 0, 1, 2, 17)) + #adjust shape values
178   theme_minimal() #clean theme
179 dev.off()
180
181 ###FILTRANDO VETORES PRINCIPAIS
182 var_coord <- pca$var$coord[, 1:2]
183 keep_vars <- apply(abs(var_coord) > 0.5, 1, any)
184 keep_vars
185 vars_to_keep <- rownames(var_coord)[keep_vars]
186 vars_to_keep
187 # Create a filtered PCA object
188 pca_filt <- pca
189 pca_filt$var$coord <- pca$var$coord[vars_to_keep, , drop = FALSE]
190 pca_filt$var$contrib <- pca$var$contrib[vars_to_keep, , drop = FALSE]
191 pca_filt$var$cos2 <- pca$var$cos2[vars_to_keep, , drop = FALSE]
192
193 ###FIX VARIABLES COORDINATES-----
194 var_coords <- as.data.frame(pca$var$coord)
195 # Manually adjust positions (modify values as needed)
196 #var_coords$Dim.1 <- var_coords$Dim.1 * 4.7 #shift right
197 #var_coords$Dim.2 <- var_coords$Dim.2 * 4.7 #shift up
198 #Fine tuning
199 #fix(var_coords)
200 #write.table(var_coords, "coords_var.txt")
201 var_coords <- read.table("coords_var.txt")
202 #rownames(var_coords) <-
203 # substr(rownames(var_coords), 3, nchar(rownames(var_coords))) #remove os 2 primeiros caract

```



```

204
205 ###FIX INDIVIDUALS COORDINATES----
206 ind_coords <- as.data.frame(pca$ind$coord)
207 # Manually adjust positions (modify values as needed)
208 #ind_coords$Dim.1 <- ind_coords$Dim.1 + 0 #shift right
209 #ind_coords$Dim.2 <- ind_coords$Dim.2 + 0.2 #shift up
210 #Fine tuning
211 #fix(ind_coords)
212 #write.table(ind_coords, "coords_ind.txt")
213 ind_coords <- read.table("coords_ind.txt")
214 #rownames(ind_coords) <-
215 # substr(rownames(ind_coords), 5, nchar(rownames(ind_coords))) #remove os 4 primeiros caract
216
217 #PLOT----
218
219 png("fig-hab_PCA_fviz.png", width = 11, height = 9, units = "in", res = 400)
220 fviz_pca_biplot(pca, repel = FALSE,
221                 col.var = "red", col.ind = "black",
222                 geom = "none", #remove default labels
223                 label = "none") + #remove both variable & individual labels
224 # Add manually adjusted variable labels
225   geom_text(data = var_coords, aes(x = Dim.1, y = Dim.2, label = rownames(var_coords)),
226            color = "red", size = 4) +
227 # Add manually adjusted individual labels
228   geom_text(data = ind_coords, aes(x = Dim.1, y = Dim.2, label = rownames(ind_coords)),
229            color = "black", size = 4, nudge_x = 0, nudge_y = 0) + #corrected column names
230   geom_point(aes(shape = t_grps$UA), size = 3) + #map shapes to t_grps$UA
231   scale_shape_manual(values = c(15, 16, 0, 1, 2, 17)) +
232   theme_minimal() + ggtitle(NULL) + labs(shape = "Sites") +
233   theme(legend.text = element_text(size = 14), #increase legend text size
234         legend.title = element_text(size = 16)) + #increase legend title size
235   guides(shape = guide_legend(override.aes = list(size = 4))) #increase legend symbols
236 dev.off()

1 #PCA R base----
2
3 dev.off() #apaga os graficos, se houver algum
4 rm(list=ls(all=TRUE)) #limpa a memória
5 cat("\014") #limpa o console
6
7 # Eigenvalues & variance explained
8 eig.val <- get_eigenvalue(pca)
9 eig_table <- data.frame(

```

```

10  PC = rownames(eig.val),
11  Eigenvalue = eig.val[, "eigenvalue"],
12  Variance_percent = round(eig.val[, "variance.percent"], 2),
13  Cumulative_variance_percent = round(eig.val[, "cumulative.variance.percent"], 2),
14  row.names = NULL)
15  eig_table
16
17  # Variable loadings (correlations) for retained PCs
18  n_pc <- 2 #number of PCs
19  loadings <- pca$var$coord[, 1:n_pc]
20
21  loadings_table <- data.frame(
22    Variable = rownames(loadings),
23    round(loadings, 3),
24    row.names = NULL)
25  colnames(loadings_table)[-1] <- paste0("PC", 1:n_pc)
26  loadings_table
27
28  # Variable contributions (%) to each PC
29  contrib <- pca$var$contrib[, 1:n_pc]
30  contrib_table <- data.frame(
31    Variable = rownames(contrib),
32    round(contrib, 2),
33    row.names = NULL)
34  colnames(contrib_table)[-1] <- paste0("PC", 1:n_pc, "_contrib")
35  contrib_table
36
37  # Combined report-ready table
38  pca_report_table <- merge(
39    loadings_table,
40    contrib_table,
41    by = "Variable",
42    sort = FALSE)
43  rownames(pca_report_table) <- pca_report_table[, 1]
44  pca_report_table <- pca_report_table[, -1]
45  pca_report_table
46
47  # Save tables
48  sink(file = "pca_table.txt", split = T)
49  pca_report_table
50  sink()

```

Averaged nutrient concentration values varied among sampling sites (Table 1). Ammonia

ranged from 54.8 $\mu\text{g L}^{-1}$ at the reference site (Site 1) to 243.7 $\mu\text{g L}^{-1}$ at Site 6, representing the highest value recorded during the study. Elevated ammonia concentrations were also observed at Sites 4 (81.6 $\mu\text{g L}^{-1}$) and 5 (80.8 $\mu\text{g L}^{-1}$), whereas lower concentrations were detected at Sites 2 (58.9 $\mu\text{g L}^{-1}$) and 3 (72.0 $\mu\text{g L}^{-1}$). Nitrite concentrations were below detection limits at all sampling sites, whereas nitrate concentrations ranged from 5.5 to 18.8 $\mu\text{g L}^{-1}$, with the highest values recorded at Sites 4 and 5. Orthophosphate was detected only at Site 3 (1.0 $\mu\text{g L}^{-1}$), while total phosphorus varied from undetectable levels at Site 1 to 50.1 $\mu\text{g L}^{-1}$ at Site 3.

Fish community. A total of six fish species were collected under a standardized sampling effort, so species representation reflected their natural abundances in the study area. Among the families recorded, Characidae was the richest, comprising five species (*Astyanax bimaculatus*, *Astyanax fasciatus*, *Hemigrammus marginatus*, *Hemigrammus rodwayi*, and *Hemigrammus unilineatus*), whereas Crenuchidae was represented by a single species, *Characidium bimaculatum*, from which only one specimen was collected. *Hemigrammus unilineatus* was the most abundant species, representing 66.3% of all individuals analyzed, followed by *Astyanax fasciatus* (14.8%) and *Hemigrammus rodwayi* (11.2%). *Astyanax bimaculatus* accounted for 4.6% of the specimens, whereas *Hemigrammus marginatus* and *Characidium bimaculatum* represented only 2.5% and 0.5%, respectively. Since species representation in the physiological analyses reflected their natural abundances, the number of individuals available was divided for each subsequent biomarker analysis, in order to reflect the natural composition of the fish assemblage. Of the total fish collected, 118 specimens were submitted to the rhodamine B accumulation assay and 78 were used for total plasma protein and moisture content analyses (Figure 4). Given its low count, *Characidium bimaculatum* was removed from all biomarker analyses.

```

1 dev.off() #apaga os graficos, se houver algum
2 rm(list=ls(all=TRUE)) #limpa a memória
3 cat("\014") #limpa o console
4
5 getwd()
6 library(openxlsx)
7 mxr23 <- read.xlsx("D:/Elvio/OneDrive/MSS/_rebio23-mxr/mxr23_Q/data/rebio23-peixes_5-10.xlsx",
8                   rowNames = T,
9                   colNames = T,
10                  sheet = "peixes-fisio")
11 mxr23#[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
12
13 ##### ARRUMANDO OS DADOS #####
14
15 mxr23_part <- mxr23[!row.names(mxr23) %in% c("4-9-85", #NA
16                                             "1-10-7"),] #As.bim grande
17

```

```

18 mxr23_part[c("1-10-4", "1-10-5"), c("Rodamina_rfu_mg", "TH_musculo_p",
19                                     "TH_branquia_p", "Proteinas_Totais_mg"
20                                     )] <- NA
21 mxr23_part <- mxr23_part[mxr23_part$Especie != "Characidium bimaculatum", ]
22
23 write.table(mxr23_part, "mxr23_part.csv",
24             sep = ";", dec = ".", #"\t",
25             row.names = TRUE,
26             quote = TRUE,
27             append = FALSE)
28 data <- read.csv("mxr23_part.csv",
29                 sep = ";", dec = ".",
30                 row.names = 1,
31                 header = TRUE,
32                 na.strings = NA)
33
34 ##### RESUMO DOS DADOS #####
35
36 library(dplyr)
37 library(tidyr)
38
39 grouped_data <- group_by(data, Especie, Amostra_tipo)
40 summarised_data <- summarise(grouped_data, count = n())
41 species_count_table <- spread(summarised_data, Amostra_tipo, count, fill = 0)
42 print(species_count_table)
43
44 species_count_table <- species_count_table %>%
45   rowwise() %>%
46   mutate(N = sum(c_across(where(is.numeric))))
47
48 resumo <- as.data.frame(species_count_table)
49 resumo
50
51 rownames(resumo) <- resumo[,1] #tem que ser um df
52 resumo[,1] <- NULL
53
54 soma <- apply(resumo, 2, sum)
55 soma
56
57 soma_row <- as.data.frame(t(soma))
58 rownames(soma_row) <- "Soma"
59
60 resumo <- rbind(resumo, soma_row)

```

```

61  resumo
62
63  ##### RESUMO SPP POR UA #####
64
65  resumo2 <- data %>%
66    group_by(Site = Site, Especie = Especie, Amostra_tipo = Amostra_tipo) %>%
67    summarise(Count = n()) %>%
68    arrange(Site, Especie, Amostra_tipo)
69
70  resumo2 <- as.data.frame(resumo2)
71  resumo2
72
73  write.table(resumo2, file = "resumo.csv", row.names = T, sep = "\t")
74  read.table("resumo.csv", check.names = F)
75
76  resumo2_wide <- resumo2 %>%
77    pivot_wider(names_from = c(Site, Amostra_tipo), values_from = Count, values_fill = list(Count = 0))
78  resumo2_wide <- as.data.frame(resumo2_wide)
79  resumo2_wide
80
81  # Create the gt table
82  library(gt)
83  gt_table <- resumo2_wide %>%
84    gt() %>%
85    tab_header(
86      title = "Species Count by Site and Sample Type"
87    ) %>%
88    cols_label(
89      Especie = "Species"
90    ) %>%
91    fmt_number(
92      columns = everything(),
93      decimals = 0
94    ) %>%
95    cols_width(
96      everything() ~ px(100)
97    ) %>%
98    opt_table_outline()
99
100  gt_table
101
102  ##### GRÁFICO POR N #####
103

```

```

104 # Filtrar por Amostra_tipo "xx"
105 filtered_Amostra_tipo <- data %>%
106   filter(Amostra_tipo != "xx")
107 data <- filtered_Amostra_tipo
108
109 # Count occurrences of each species by Amostra_tipo and Site
110 species_count <- data %>%
111   group_by(Site, Especie, Amostra_tipo) %>%
112   summarise(Count = n(), .groups = 'drop')
113
114 # Plot the counts
115 library(ggplot2)
116
117 counts <- ggplot(species_count, aes(x = Amostra_tipo, y = Count, fill = Especie)) +
118   geom_bar(stat = "identity", position = position_dodge()) +
119   facet_wrap(
120     ~ Site,
121     labeller = labeller(
122       Site = c(
123         "B-P05" = "Site 1",
124         "B-P06" = "Site 2",
125         "B-P07" = "Site 3",
126         "B-P08" = "Site 4",
127         "B-P09" = "Site 5",
128         "B-P10" = "Site 6"
129       )
130     )
131   ) +
132   scale_x_discrete(
133     labels = c(
134       "rd" = "MXR",
135       "th" = "Moisture"
136     )
137   ) +
138   scale_fill_grey(start = 0.2, end = 0.8) +
139   labs(title = "",
140        x = "Sample use",
141        y = "Count",
142        fill = "Species") +
143   theme_minimal() +
144   theme(axis.text.x = element_text(angle = 45, hjust = 1))
145 counts
146 ggsave(counts, dpi = 300, filename = "fig-counts.png", bg = "white")

```

```

147
148
149 # Count occurrences of each species by Amostra_tipo and Site
150 species_count <- data %>%
151   group_by(Site, Espécie, Amostra_tipo) %>%
152   summarise(Count = n(), .groups = 'drop')
153 # Plot the counts with species on the x-axis
154 ggplot(species_count, aes(x = Espécie, y = Count, fill = Amostra_tipo)) +
155   geom_bar(stat = "identity", position = position_dodge()) +
156   facet_wrap(~ Site) +
157   labs(title = "Count of Each Species per Amostra_tipo for Each Site",
158        x = "Species",
159        y = "Count") +
160   theme_minimal() +
161   theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

Plasmatic protein dosage. The number of plasmatic protein dosages (n=44) was smaller than the total number of fish collected for this part of the analysis (n=78), because of the difficulty on collecting blood from smaller specimens and the need to collect a minimum amount of plasma (Figure 5).

```

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",
7                 row.names = 1,
8                 header = TRUE,
9                 na.strings = NA)
10
11 table(data$Espécie, is.na(data$Proteinas_Totais_mg))
12
13 ##### GRAFICO POR ANÁLISE #####
14
15 #N #data <- data %>% mutate(N = 1)
16 #RPC
17 #Rodamina_rfu_mg
18 #TH_musculo_p
19 #TH_branquias_p
20 #Proteinas_Totais_mg
21
22 # Convert the column to numeric, replace commas with dots for proper conversion

```

```

23 data$Proteinas_Totais_mg <- as.numeric(gsub(",", ".", data$Proteinas_Totais_mg))
24 VAR <- deparse(substitute(data$Proteinas_Totais_mg))
25 var <- "Proteinas_Totais_mg" %>% rlang::sym()
26 str(data)
27
28 # Calculate the average for each combination of species, Amostra_tipo, and Site
29 average <- summarise(
30   group_by(data, Site, Espécie, Amostra_tipo),
31   Average = mean(!var, na.rm = TRUE),
32   n = sum(!is.na(!var)),
33   SE = ifelse(
34     sum(!is.na(!var)) > 1,
35     sd(!var, na.rm = TRUE) / sqrt(sum(!is.na(!var))),
36     0
37   ),
38   .groups = "drop"
39 )
40 average <- average[
41   average$Amostra_tipo == "th" & !is.nan(average$Average),
42 ]
43 # Plot the average
44 ggplot(average, aes(x = Espécie, y = Average, fill = Amostra_tipo)) +
45   geom_bar(stat = "identity", position = position_dodge()) +
46   facet_wrap(~ Site) +
47   labs(title = paste("Average POR ANÁLISE para", VAR, "per Species and Amostra_tipo for Each",
48     x = "Species",
49     y = paste("Average", VAR)) +
50   theme_minimal() +
51   theme(axis.text.x = element_text(angle = 45, hjust = 1))
52
53 ##### PLOT FINAL #####
54
55 prot_tots <-
56   ggplot(average, aes(x = Espécie, y = Average, fill = Amostra_tipo)) +
57   geom_bar(stat = "identity", position = position_dodge(), fill = "grey70") +
58   geom_errorbar(
59     aes(
60       ymin = Average - SE,
61       ymax = Average + SE
62     ),
63     width = 0.2
64   ) +
65   geom_text(

```



```

66     aes(
67       y = 10,
68       label = paste0("n=", n)
69     ),
70     size = 3
71   ) +
72   facet_wrap(
73     ~ Site,
74     labeller = labeller(
75       Site = c(
76         "B-P05" = "Site 1",
77         "B-P06" = "Site 2",
78         "B-P07" = "Site 3",
79         "B-P08" = "Site 4",
80         "B-P09" = "Site 5",
81         "B-P10" = "Site 6"
82       )
83     )
84   ) +
85   scale_fill_grey(start = 0.2, end = 0.8) +
86   labs(title = "",
87        x = "Species",
88        y = "Plasmatic protein dosage (mg)",
89        fill = "Species") +
90   theme_grey() +
91   theme(axis.text.x = element_text(angle = 45, hjust = 1),
92         legend.position = "none")
93 prot_tots
94 ggsave(prot_tots, dpi = 300, filename = "fig-prot_tots.png", bg = "white")

1  ##### GRÁFICO POR ESPÉCIE #####
2
3  # Filter out the species Astyanax bimaculatus and Astyanax fasciatus
4
5  filtered_data <- data
6
7  filtered_data <- data %>%
8    filter(!Especie %in% c("Astyanax bimaculatus",
9                          "Astyanax fasciatus",
10                         "Characidium bimaculatum")) #EXCLUI USANDO O "!"
11 filtered_data <- data %>%
12   filter(Especie %in% c("Hemigrammus unilineatus")) #ESCOLHE
13

```

```

14 # Calculate the average for each combination of Amostra_tipo, Espécie, and Site
15 average <- filtered_data %>%
16   group_by(Site, Amostra_tipo, Espécie) %>%
17   summarise(Average = mean(!var, na.rm = TRUE), .groups = 'drop')
18 # Plot the average RPC with Amostra_tipo on the x-axis
19 ggplot(average, aes(x = Amostra_tipo, y = Average, fill = Espécie)) +
20   geom_bar(stat = "identity", position = position_dodge()) +
21   facet_wrap(~ Site) +
22   labs(title = paste("Average", VAR, "per Amostra_tipo for Each Site"),
23        x = "Amostra Tipo",
24        y = paste("Average", VAR)) +
25   theme_minimal() +
26   theme(axis.text.x = element_text(angle = 45, hjust = 1))
27
28 ##### CORRELAÇÕES #####
29
30 data <- filtered_data
31
32 colnames(data)
33 cor <- data %>%
34   select(Ponto, Ref_site, Compr.total_mm, Compr.padrao_mm, Altura_mm,
35          Peso_g, RPC, Rodamina_rfu_mg, TH_branquia_p, TH_musculo_p,
36          Proteinas_Totais_mg)
37 str(cor)
38 cor[] <- lapply(cor, function(x) as.numeric(gsub(",", ".", x)))
39
40 plot(cor)
41 plot(cor[,3:7])
42
43 cor(cor,method="pearson")
44 cor(cor[,1:3], method="spearman")
45
46 library(psych)
47 pairs.panels(cor[,8:11],
48              method = "pearson", # correlation method
49              scale = FALSE, lm = FALSE,
50              hist.col = "#00AFBB", pch = 19,
51              density = TRUE, # show density plots
52              ellipses = TRUE, # show correlation ellipses
53              alpha = 0.5
54 )
55
56 cor %>%

```

```

57   with(cor.test(RPC, Peso_g, method = "pearson"))
58
59   with(cor, plot(scale(RPC), scale(Peso_g)))
60   with(cor, plot(RPC, Peso_g))
61
62   plot((m_part$"Overh_veg" - mean(m_part$"Overh_veg")) / sd(m_part$"Overh_veg"))
63   #plot((m_trns$"m.elev" - mean(m_trns$"m.elev")) / sd(m_trns$"m.elev"))
64   with(cor, plot((RPC - mean(Peso_g)) / sd(Peso_g)))
65   with(cor, plot((RPC - mean(RPC)) / sd(RPC),
66                 (Peso_g - mean(Peso_g)) / sd(Peso_g)))

```

Plasma total protein concentrations differed between sampling sites when compared with the reference site (Site 1). Mean protein concentrations were significantly higher at the sites 4, 5 and 6 (91.25 ± 5.68 , 74.22 ± 3.31 and 74.79 ± 6.55 mg, respectively) than the reference site (33.53 ± 0.32 mg; Wilcoxon rank-sum test, $p < 0.02$). In contrast, no significant increase was observed at Sites 2 and 3 (29.41 ± 1.59 and 33.75 ± 0.80 mg, respectively; $p > 0.4$). The highest mean protein concentration was recorded at Site 4, representing an approximately 2.7-fold increase relative to the reference site.

```

1  dev.off()
2  rm(list=ls(all=TRUE))
3  cat("\014")
4
5  data <- read.csv("mxr23_part.csv",
6                  sep = ";", dec = ".",
7                  row.names = 1,
8                  header = TRUE,
9                  na.strings = NA)
10
11  table(data$Especie, is.na(data$Proteinas_Totais_mg))
12  var <- "Proteinas_Totais_mg"
13
14  # Reference site
15  ref <- data[[var]][
16    data$Site == "B-P05" &
17    !is.na(data[[var])]
18  ]
19
20  # Sites to compare
21  sites <- setdiff(unique(data$Site), "B-P05")
22
23  # Run tests
24  results <- data.frame()

```

```

25
26 for(site in sites){
27
28     test_data <- data[[var]][
29     data$Site == site &
30     !is.na(data[[var])]
31 ]
32
33 tt <- t.test(
34     test_data,
35     ref,
36     alternative = "greater"
37 )
38
39 results <- rbind(
40     results,
41     data.frame(
42         Site = site,
43         Mean_Site = mean(test_data),
44         SD_Site = sd(test_data),
45         SE_Site = sd(test_data) / sqrt(length(test_data)),
46         Mean_Ref = mean(ref),
47         SD_Ref = sd(ref),
48         SE_Ref = sd(ref) / sqrt(length(ref)),
49         N_Site = length(test_data),
50         N_Ref = length(ref),
51         t = unname(tt$statistic),
52         df = unname(tt$parameter),
53         p = tt$p.value
54     )
55 )
56 }
57
58 results
59
60 results$Significant <- ifelse(results$p < 0.05, "Yes", "No")
61
62 results$p <- ifelse(
63     results$p < 0.001,
64     "<0.001",
65     sprintf("%.3f", results$p)
66 )
67 #sink("t-tests_prot_tots.txt", append = TRUE)

```

```

68 results
69 #sink()
70
71 ##### NÃO-PARAMETRICO #####
72
73 results_w <- data.frame()
74
75 for(site in sites){
76
77     test_data <- data[[var]][
78         data$Site == site &
79         !is.na(data[[var]])
80     ]
81
82     wt <- wilcox.test(
83         test_data,
84         ref,
85         alternative = "greater"
86     )
87
88     results_w <- rbind(
89         results_w,
90         data.frame(
91             Site = site,
92             W = unname(wt$statistic),
93             p = wt$p.value
94         )
95     )
96 }
97
98 results_w$Significant <- ifelse(results_w$p < 0.05, "Yes", "No")
99
100 results_w$p <- ifelse(
101     results_w$p < 0.001,
102     "<0.001",
103     sprintf("%.3f", results_w$p)
104 )
105
106 #sink("t-tests_np.txt", append = FALSE)
107 results_w
108 #sink()

```

Moisture content. Gill and muscle tissues moisture content showed similar results, with gill

moisture concentrations being significantly higher than those observed at the reference site (Site 1; $82.82 \pm 1.75\%$) at Sites 5 and 6 (mean values of $90.13 \pm 1.19\%$ and $93.89 \pm 0.98\%$, Wilcoxon rank-sum test; $p < 0.004$). No significant increases were detected at Sites 2, 3, or 4 ($84.35 \pm 1.60\%$, $85.63 \pm 1.94\%$, and $82.84 \pm 0.75\%$, respectively; $p > 0.1$ in all cases).

The muscle moisture concentrations at the reference site averaged $80.31 \pm 1.48\%$. Significantly higher values were observed at Sites 4, 5 and 6, where mean moisture concentrations reached $82.38 \pm 0.62\%$ and $84.29 \pm 0.88\%$, $86.02 \pm 2.49\%$, respectively (Wilcoxon rank-sum test; $p = 0.04$). In contrast, moisture concentrations at Sites 2 and 3 ($80.11 \pm 0.66\%$ and $80.58 \pm 0.94\%$) did not differ significantly from those at the reference site ($p > 0.39$) (Figure 6).

```

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",
7                 row.names = 1,
8                 header = TRUE,
9                 na.strings = NA)
10
11 table(data$Especie, is.na(data$TH_branquia_p))
12 table(data$Especie, is.na(data$TH_musculo_p))
13
14 ##### GRAFICO POR ANÁLISE #####
15
16 #N #data <- data %>% mutate(N = 1)
17 #RPC
18 #Rodamina_rfu_mg
19 #TH_musculo_p
20 #TH_branquias_p
21 #Proteinas_Totais_mg
22
23 library(dplyr)
24 library(tidyr)
25 library(ggplot2)
26
27 # Convert the column to numeric, replace commas with dots for proper conversion
28 data$TH_branquia_p <- as.numeric(gsub(",", ".", data$TH_branquia_p))
29 data$TH_musculo_p <- as.numeric(gsub(",", ".", data$TH_musculo_p))
30 str(data)
31
32 # Long format

```

```

33 th_data <- pivot_longer(
34   data,
35   cols = c(TH_branquia_p, TH_musculo_p),
36   names_to = "Tecido",
37   values_to = "TH"
38 )
39
40 # Calculate the average for each combination of species, Amostra_tipo, and Site
41 average <- summarise(
42   group_by(th_data, Site, Especie, Tecido),
43   Average = mean(TH, na.rm = TRUE),
44   n = sum(!is.na(TH)),
45   SE = ifelse(
46     n > 1,
47     sd(TH, na.rm = TRUE) / sqrt(n),
48     0
49   ),
50   .groups = "drop"
51 )
52 # Remove groups with no observations
53 average <- average[average$n > 0, ]
54
55 ##### PLOT FINAL #####
56
57 prot_th <-
58   ggplot(
59     average,
60     aes(
61       x = Especie,
62       y = Average,
63       fill = Tecido
64     )
65   ) +
66   geom_col(
67     position = position_dodge(width = 0.9)
68   ) +
69   geom_errorbar(
70     aes(
71       ymin = Average - SE,
72       ymax = Average + SE
73     ),
74     width = 0.2,
75     position = position_dodge(width = 0.9)

```

```

76   ) +
77   geom_text(
78     aes(
79       y = 5,
80       label = paste0("", n)
81     ),
82     size = 3,
83     position = position_dodge(width = 0.9)
84   ) +
85   facet_wrap(
86     ~Site,
87     labeller = labeller(
88       Site = c(
89         "B-P05" = "Site 1",
90         "B-P06" = "Site 2",
91         "B-P07" = "Site 3",
92         "B-P08" = "Site 4",
93         "B-P09" = "Site 5",
94         "B-P10" = "Site 6"
95       )
96     )
97   ) +
98   scale_fill_grey(
99     start = 0.4,
100    end = 0.8,
101    labels = c(
102      "TH_branquia_p" = "Gill",
103      "TH_musculo_p" = "Muscle"
104    )
105  ) +
106  labs(
107    x = "Species",
108    y = "Moisture concentration (%)",
109    fill = "Tissue"
110  ) +
111  theme_grey() +
112  theme(
113    axis.text.x = element_text(
114      angle = 45,
115      hjust = 1
116    ),
117    strip.background = element_rect(
118      fill = "grey85",

```



```

119     colour = "grey85"
120   )
121 )
122 prot_th
123 ggsave(prot_th, dpi = 300, filename = "fig-prot_th.png", bg = "white")

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",
7                 row.names = 1,
8                 header = TRUE,
9                 na.strings = NA)
10
11 table(data$Especie, is.na(data$TH_branquia_p))
12 table(data$Especie, is.na(data$TH_musculo_p))
13
14 var <- "TH_musculo_p"
15
16 # Reference site
17 ref <- data[[var]][
18   data$Site == "B-P05" &
19   !is.na(data[[var])]
20 ]
21
22 # Sites to compare
23 sites <- setdiff(unique(data$Site), "B-P05")
24
25 # Run tests
26 results <- data.frame()
27
28 for(site in sites){
29
30   test_data <- data[[var]][
31     data$Site == site &
32     !is.na(data[[var])]
33   ]
34
35   tt <- t.test(
36     test_data,
37     ref,

```

```

38     alternative = "greater"
39   )
40
41   results <- rbind(
42     results,
43     data.frame(
44       Site = site,
45       Mean_Site = mean(test_data),
46       SD_Site = sd(test_data),
47       SE_Site = sd(test_data) / sqrt(length(test_data)),
48       Mean_Ref = mean(ref),
49       SD_Ref = sd(ref),
50       SE_Ref = sd(ref) / sqrt(length(ref)),
51       N_Site = length(test_data),
52       N_Ref = length(ref),
53       t = unname(tt$statistic),
54       df = unname(tt$parameter),
55       p = tt$p.value
56     )
57   )
58 }
59
60 results
61
62 results$Significant <- ifelse(results$p < 0.05, "Yes", "No")
63
64 results$p <- ifelse(
65   results$p < 0.001,
66   "<0.001",
67   sprintf("%.3f", results$p)
68 )
69 #sink("t-tests_prot_th.txt", append = TRUE)
70 results
71 #sink()
72
73 ##### NÃO-PARAMETRICO #####
74
75 results_w <- data.frame()
76
77 for(site in sites){
78
79   test_data <- data[[var]][
80     data$Site == site &

```

```

81     !is.na(data[[var]])
82   ]
83
84   wt <- wilcox.test(
85     test_data,
86     ref,
87     alternative = "greater"
88   )
89
90   results_w <- rbind(
91     results_w,
92     data.frame(
93       Site = site,
94       W = unname(wt$statistic),
95       p = wt$p.value
96     )
97   )
98 }
99
100 results_w$Significant <- ifelse(results_w$p < 0.05, "Yes", "No")
101
102 results_w$p <- ifelse(
103   results_w$p < 0.001,
104   "<0.001",
105   sprintf("%.3f", results_w$p)
106 )
107
108 #sink("t-tests_np.txt", append = TRUE)
109 results_w
110 #sink()

```

MXR phenotype activity. Rhodamine B concentrations differed among sampling sites relative to the reference site (Site 1), which exhibited a mean concentration of $38,396.35 \pm 4,749.86$ RFU mg⁻¹. Significantly higher concentrations were observed at Sites 4, 5 and 6, with mean values of $73,113.69 \pm 5,311.65$, $67,116.05 \pm 7,023.54$ and $62,929.41 \pm 6,196.21$ RFU mg⁻¹, respectively (Wilcoxon rank-sum test; $p < 0.01$). In contrast, Rhodamine B concentrations at Sites 2 and 3 ($16,978.11 \pm 2,139.98$ and $20,973.12 \pm 2,373.55$ RFU mg⁻¹, respectively) were actually significantly lower than those observed at the reference site (two-sided Wilcoxon rank-sum test; $p < 0.01$). The highest mean concentration was recorded at Site 4, corresponding to an approximately 1.9-fold increase relative to the reference site.

```

1 dev.off()
2 rm(list=ls(all=TRUE))

```

```

3  cat("\014")
4
5  data <- read.csv("mxr23_part.csv",
6                  sep = ";", dec = ".",
7                  row.names = 1,
8                  header = TRUE,
9                  na.strings = NA)
10
11 table(data$Especie, is.na(data$Rodamina_rfu_mg))
12
13 ##### GRAFICO POR ANÁLISE #####
14
15 #N #data <- data %>% mutate(N = 1)
16 #RPC
17 #Rodamina_rfu_mg
18 #TH_musculo_p
19 #TH_branquias_p
20 #Proteinas_Totais_mg
21
22 # Convert the column to numeric, replace commas with dots for proper conversion
23 data$Rodamina_rfu_mg <- as.numeric(gsub(",", ".", data$Rodamina_rfu_mg))
24 VAR <- deparse(substitute(data$Rodamina_rfu_mg))
25 var <- "Rodamina_rfu_mg" %>% rlang::sym()
26 str(data)
27
28 # Calculate the average for each combination of species, Amostra_tipo, and Site
29 average <- summarise(
30   group_by(data, Site, Especie, Amostra_tipo),
31   Average = mean(!var, na.rm = TRUE),
32   n = sum(!is.na(!var)),
33   SE = ifelse(
34     sum(!is.na(!var)) > 1,
35     sd(!var, na.rm = TRUE) / sqrt(sum(!is.na(!var))),
36     0
37   ),
38   .groups = "drop"
39 )
40 average <- average[
41   average$Amostra_tipo == "rd" & !is.nan(average$Average),
42 ]
43 # Plot the average
44 ggplot(average, aes(x = Especie, y = Average, fill = Amostra_tipo)) +
45   geom_bar(stat = "identity", position = position_dodge()) +

```

```

46 facet_wrap(~ Site) +
47 labs(title = paste("Average POR ANÁLISE para", VAR, "per Species and Amostra_tipo for Each
48       x = "Species",
49       y = paste("Average", VAR)) +
50 theme_minimal() +
51 theme(axis.text.x = element_text(angle = 45, hjust = 1))
52
53 ##### PLOT FINAL #####
54
55 rodam <-
56 ggplot(average, aes(x = Especie, y = Average, fill = Amostra_tipo)) +
57 geom_bar(stat = "identity", position = position_dodge(), fill = "grey70") +
58   geom_errorbar(
59     aes(
60       ymin = Average - SE,
61       ymax = Average + SE
62     ),
63     width = 0.2
64   ) +
65   geom_text(
66     aes(
67       y = 5000,
68       label = paste0("n=", n)
69     ),
70     size = 3
71   ) +
72   facet_wrap(
73     ~ Site,
74     labeller = labeller(
75       Site = c(
76         "B-P05" = "Site 1",
77         "B-P06" = "Site 2",
78         "B-P07" = "Site 3",
79         "B-P08" = "Site 4",
80         "B-P09" = "Site 5",
81         "B-P10" = "Site 6"
82       )
83     )
84   ) +
85   scale_fill_grey(start = 0.2, end = 0.8) +
86   labs(title = "",
87        x = "Species",
88        y = "Rhodamine B concentration (RFU/mg tissue)",

```

```

89     fill = "Species") +
90     theme_grey() +
91     theme(axis.text.x = element_text(angle = 45, hjust = 1),
92           legend.position = "none")
93 rodam
94 ggsave(rodam, dpi = 300, filename = "fig-rodam.png", bg = "white")

1  dev.off()
2  rm(list=ls(all=TRUE))
3  cat("\014")
4
5  data <- read.csv("mxr23_part.csv",
6                  sep = ";", dec = ".",
7                  row.names = 1,
8                  header = TRUE,
9                  na.strings = NA)
10
11 table(data$Especie, is.na(data$Rodamina_rfu_mg))
12 var <- "Rodamina_rfu_mg"
13
14 # Reference site
15 ref <- data[[var]][
16   data$Site == "B-P05" &
17   !is.na(data[[var])]
18 ]
19
20 # Sites to compare
21 sites <- setdiff(unique(data$Site), "B-P05")
22
23 # Run tests
24 results <- data.frame()
25
26 for(site in sites){
27
28   test_data <- data[[var]][
29     data$Site == site &
30     !is.na(data[[var])]
31 ]
32
33   tt <- t.test(
34     test_data,
35     ref,
36     alternative = "greater"

```

```

37   )
38
39   results <- rbind(
40   results,
41   data.frame(
42     Site = site,
43     Mean_Site = mean(test_data),
44     SD_Site = sd(test_data),
45     SE_Site = sd(test_data) / sqrt(length(test_data)),
46     Mean_Ref = mean(ref),
47     SD_Ref = sd(ref),
48     SE_Ref = sd(ref) / sqrt(length(ref)),
49     N_Site = length(test_data),
50     N_Ref = length(ref),
51     t = unname(tt$statistic),
52     df = unname(tt$parameter),
53     p = tt$p.value
54   )
55 )
56 }
57
58 results
59
60 results$Significant <- ifelse(results$p < 0.05, "Yes", "No")
61
62 results$p <- ifelse(
63   results$p < 0.001,
64   "<0.001",
65   sprintf("%.3f", results$p)
66 )
67 #sink("t-tests_rodam.txt", append = TRUE)
68 results
69 #sink()
70
71 ##### NÃO-PARAMETRICO #####
72
73 results_w <- data.frame()
74
75 for(site in sites){
76
77   test_data <- data[[var]][
78     data$Site == site &
79     !is.na(data[[var])]

```

```

80   ]
81
82   wt <- wilcox.test(
83     test_data,
84     ref,
85     alternative = "two.sided"
86   )
87
88   results_w <- rbind(
89     results_w,
90     data.frame(
91       Site = site,
92       W = unname(wt$statistic),
93       p = wt$p.value
94     )
95   )
96 }
97
98 results_w$Significant <- ifelse(results_w$p < 0.05, "Yes", "No")
99
100 results_w$p <- ifelse(
101   results_w$p < 0.001,
102   "<0.001",
103   sprintf("%.3f", results_w$p)
104 )
105
106 #sink("t-tests_np.txt", append = TRUE)
107 results_w
108 #sink()

```

4 Discussion

The present study evaluated whether biomarkers of physiological stress in headwater stream fish could detect environmental disturbances occurring within and around a protected area of Atlantic Forest in northeastern Brazil. Despite sites within the administrative boundaries of the REBIO showing no signs alteration, from both environmental variables and biomarkers, significant differences in biomarker responses were observed among sampling sites, indicating that environmental conditions are not homogeneous throughout the stream studied. Elevated levels of all biomarkers at sites located outside or near the boundaries of the protected area suggest exposure to environmental stressors potentially associated with anthropogenic activities. These findings support the use of biochemical and physiological biomarkers as sensitive

tools for detecting sublethal effects of environmental degradation, even in landscapes that contain legally protected areas.

stress levels occurring inside and outside of a Conservation Unity was tested in the Guaribas Environmental Reserve (REBIO Guaribas), one of the 22 CU's inside the Atlantic forest area of Paraíba state, located in the Zona da Mata, wettest and most economically relevant mesoregion from the state (Figure 1). Grouped, according to SNUC, as one of the integral protection areas (aimed to conserve biodiversity, being minority within the biogeographical province – 33,45% of the CU's), the definition of environmental reserves prohibits human interference inside their limits, except for management actions that aim its preservation and recovery. Despite being one of the most restricted types of Conservation Units, there are 22 populations that influence REBIO Guaribas due to historic factors of occupation and absence of an effective buffer zone around the three areas that make up the reserve. Those populations are mainly rural and cause direct impact, by using wood and plants from the reserve, and indirect, causing siltation of rivers and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil, promoting possible comparable ecophysiological responses in the ichthyofauna inside and around the CU (Arruda et al., 2013; MMA, 2026).

The ecophysiological evaluation of human impact on the forest, which is close to areas of monoculture and natural vegetation modified for livestock, was mainly guided by the fish species inventory of REBIO Guaribas made by Gouveia et al. (2017), in a study that lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012) - and none are listed as threatened. The predominance of small fish inhabiting small basins explains the higher level of endemism in streams, resulting from the allopatric speciation generated by the low dispersion made by these fish and the consequent isolation of populations (Gouveia et al., 2017). Having in mind that these stream fish are phylogenetic close, this paper offers representative data for the entire local ichthyofauna using species of *Hemigrammus* genus, of the Characiformes order, characterized by a black conspicuous spot delimited on the dorsal fin. Its type species, *Hemigrammus unilineatus*, has a compressed and slightly elongated body and exhibits wide geographical distribution both in Central and South America, with registers in other Brazilian states, such as Alagoas and Pernambuco.

Biomarkers. The increase of total plasmatic protein concentration in the animals exposed to pollution is consistent with the (Sabae & Mohamed, 2015) study, which attributes this increase to five possibilities: activation of metabolic systems as answer to pollutants exposure, degradation of cellular material in the liver, severe pathological conditions in the liver and kidneys, loss of water in the plasma, and induction of proteic synthesis in the liver. The average values of moisture content, parameter related to the osmoregulatory function, were 84,3% for the gill and 80,3% for the muscle of fish inside the REBIO Guaribas (n=38 fish). Regarding the fish from sampling sites surrounding the reserve (n=40), the averages were 88,9% for the

gills and 82,3% for the muscle, increasing, in order, only about 5,6% and 2,5% the values from inside, results that are not very expressive, like the ones described by Haredi et al. (2020). The average values of moisture content, parameter related to the osmoregulatory function, were 84,3% for the gill and 80,3% for the muscle of fish inside the REBIO Guaribas (n=38 fish). Regarding the fish from sampling sites surrounding the reserve (n=40), the averages were 88,9% for the gills and 82,3% for the muscle, increasing, in order, only about 5,6% and 2,5% the values from inside, results that are not very expressive, like the ones described by Haredi et al. (2020). The overall statistical analysis performed for moisture content reinforces this as a more discreet and inefficient parameter to make direct quantitative comparison for animals exposed to low salinity variation from the environment, but still accurate in determining the uniformity degree of more representative groups (i.e., fish from inside and outside the Conservative Unity). The average values of normalized fluorescence measured in fish gills were 25.449 rfu/mg (relative fluorescence units/milligramme of tissue) for inside the Conservative Unity (n=54 fish) and 67.138,3 rfu/mg for outside (n=57), value nearly 2,64 times bigger. This difference between averages is opposite to the expected (higher fluorescence signals a lower MXR phenotype activity, meaning a lower exposition to pollutants), that can be explained by the presence of MXR phenotype inhibitors, which were highly correlated to more polluted waters by Kurelec et al. (2000). Among the possible pollutants presents, pesticides, fragrances, microbial degradation products and natural inhibitors from invasive species were described as compounds with high affinity to the Pgp or inhibiting the PKC (protein kinase C) modulator, resulting in high accumulation of rhodamine B (Smital et al., 2004) Despite the unexpected result, the variance difference between data from the MXR phenotype activity inside and outside the REBIO Guaribas (confirmed by the Kruskal-Wallis test and illustrated in *The abiotic data from every sampling site water was measured and reunited in Tab. 1. The abnormal proportion of ammonia indicates the presence of effluents in the water outside the Reserva Biológica Guaribas, and, as discussed by (Piedras et al., 2006), this elevated concentration generates metabolic ammonia retention, causing toxicity. The same paper also points out that the toxicity of non-ionized ammonia in the aquatic environment is greater for smaller organisms, but depends on the interaction with other environmental variables, such as temperature, pH and salinity, demanding specific studies with the target species to determine their tolerance to ammonia. The Pearson correlation analysis among biotic data showed only one significant correlation, between Total Protein and MXR phenotype activity (0,6 - moderate and positive), and among abiotic data only two significance, between Nitrate e Phosphor (0,69 - moderate and positive) and between Dissolved Oxygen and Ammonia (-0,82 - strong and negative). It was also detected significant correlation among both data, between Gill Moisture Content and Ammonia (0,55 - moderate and positive), between Gill Moisture Content and Dissolved Oxygen (-0,62 - moderate and negative), and between MXR activity and Nitrate (-0,62 - moderate and negative).* The elevated ammonia concentrations observed at Sites 4–6, particularly at Site 6, may indicate inputs of domestic or agricultural effluents into the aquatic environment. According to Piedras et al. (2006), increased environmental ammonia can lead to metabolic ammonia retention and toxic effects in fish, although toxicity depends on interactions with other environmental variables such as temperature, pH, and salinity. The strong negative correlation between dissolved oxygen and ammonia concentration observed in

the present study further suggests that sites with greater organic inputs may be experiencing water quality degradation.

All parameters were effective to point out statistical differences between data from inside the REBIO Guaribas and its surroundings, highlighting - alongside the higher levels of ammonia in the water and plasmatic proteins in the fish - that changes in the environment are impacting the fish from the streams studied. Beyond that, the statistical analysis of physiological parameters from fish collected inside the Conservation Unity showed uniform results among its three sampling sites, consistent with what is expected from natural environments with little human influence. The same did not occur among the three sites outside the CU (with the exception of the MXR phenotype activity), indicating the action of different levels of environmental stress. The differences between the average results from the physiological parameters measured inside and outside the Conservation Unity were illustrated in the fig-bar, highlighting the total protein as one that proved itself as an excellent biomarker to detect pollutants in the aquatic environment.

These preliminary obtained data were precise to indicate significant differences among the conditions in which the fish inhabited. But as much MXR phenotype activity data were as disparate as the ones of total protein and even more precise to point out the uniformity among sampling sites inside and outside the REBIO Guaribas, it showed values opposite to what was expected, with higher fluorescence in the tissue of fish outside the reserve, normally indicating lower MXR phenotype activity. This inconsistency can be explained by the possible presence of chemosensitizers in the water, since the fish did not went through a decontamination period, and thus, pollutants could still be bound to the active Pgp site and preventing rhodamine B to attach to it. The presence of such chemosensitizers can be quantified through the creation of a calibration curve, as described by (Kurelec et al., 2000), but, until done, the relevance of the MXR phenotype analysis must be restricted to the statistical context.

From the final results, an important positive evaluation can be made regarding the environmental protection that the Conservative Unity does within its extension, even when analysing one with historical problems of delimitation and still not entirely free of anthropic impacts. Adding the fact that it is located within the Atlantic forest, classified as a Hotspot, the verification of the efficiency of a CU in this biogeographical province is even more relevant, reinforcing the need for governments to create more legally protected areas as a key strategy to reduce global ecological crisis. This study also made an important description on the use of physiological biomarkers in ecological analysis, raising ecophysiology potential as a basis to future assays, which can be incorporated into effectiveness index such as seen in (Masullo et al., 2020) paper and presenting quantifiable data that can complement studies with analysis of historical and geographical data, such as the one from (Assis et al., 2021).

Conclusions

Acknowledgements

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Adamastor Coutinho Pinto: Investigation, Formal analysis, Writing – original draft.

Larissa Rafaela Caetano da Silva: Data curation, Formal analysis.

Mayara Mirelly da Silva Monteiro: Data curation, Formal analysis, Software, Writing – review & editing.

Alice da Silva Barros: Data curation, Formal analysis, Software, Writing – review & editing.

Enelise Marcelle Amado: Data curation, Methodology, Resources, Supervision, Validation, Writing – review & editing.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Ethical approval

Fish collection was authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license SISBIO-89415-2) and the project was approved by the Ethics Committee on Animal Use of the State University of Paraíba (protocol CEUA-048/2023).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and fluency of the writing. The authors carefully reviewed and revised all AI-generated suggestions and take full responsibility for the accuracy and content of the final manuscript (CNPq, 2026).

References

- Abilhoa, V., R., R., Bornatowski, H., & S. Vitule, J. R. (2011). *Fishes of the Atlantic Rain Forest Streams: Ecological Patterns and Conservation* (O. Grillo, Ed.). InTech. <https://doi.org/10.5772/24540>
- Amoatey, P., & Baawain, M. S. (2019). Effects of pollution on freshwater aquatic organisms. *Water Environment Research*, 91(10), 1272–1287. <https://doi.org/10.1002/wer.1221>
- APHA. (1998). *Standard methods for the examination of water and wastewater*. <https://www.standardmethods.org/>
- Arruda, D. B., Cunha, B. P., & Rêgo, K. M. C. (2013). Conflitos entre ReBio guaribas e comunidades locais: (in)justiça ambiental e ecologia política. *Revista Direitos Emergentes na Sociedade Global*, 2(2).
- Assis, P., Faria, K., & Bayer, M. (2021). Unidades de conservação e sua efetividade na proteção dos recursos hídricos na bacia do rio araguaia. *Sociedade & Natureza*, 34. <https://doi.org/10.14393/SN-v34-2022-60335>
- Ayrapetyan, S. (2012). Cell hydration as a universal marker for detection of environmental pollution. *The Environmentalist*, 32(2), 210–221. <https://doi.org/10.1007/s10669-011-9380-3>
- Bard, S. (2000). Multixenobiotic resistance as a cellular defense mechanism in aquatic organisms. *Aquatic Toxicology*, 48(4), 357–389. [https://doi.org/10.1016/S0166-445X\(00\)00088-6](https://doi.org/10.1016/S0166-445X(00)00088-6)
- Bradford, M. M. (1976). *A Rapid and Sensitive Method for the Quantitation of Microgram Quantities of Protein Utilizing the Principle of Protein-Dye Binding*.
- BRASIL. (2000). *Lei no 9.985, de 18 de julho de 2000. Regulamenta o art. 225, § 1o, incisos i, II, III e VII da constituição federal, institui o sistema nacional de unidades de conservação da natureza e dá outras providências*. https://www.planalto.gov.br/ccivil_03/leis/19985.htm
- Cazenave, J., Bacchetta, C., Parma, M. J., Scarabotti, P. A., & Wunderlin, D. A. (2009). Multiple biomarkers responses in *Prochilodus lineatus* allowed assessing changes in the water quality of Salado River basin (Santa Fe, Argentina). *Environmental Pollution*, 157(11), 3025–3033. <https://doi.org/10.1016/j.envpol.2009.05.055>

- CNPq. (2026). *Portaria 2664/2026 de 6 de março de 2026 - política de integridade na atividade científica do CNPq*. http://memoria2.cnpq.br/web/guest/view/-/journal_content/56_INSTANCE_0oED/10157/23142775
- Costa, D. A., Azevedo, J. P. S., Santos, M. A., & Assumpção, R. S. F. V. (2020). Water quality assessment based on multivariate statistics and water quality index of a strategic river in the Brazilian Atlantic Forest. *Scientific Reports*, 10(1), 22038. <https://doi.org/10.1038/s41598-020-78563-0>
- CRedit. (2026). CRediT - contributor role taxonomy. *NISO/CRediT Standing Committee*. <https://credit.niso.org/>
- David, D. D., Lima, O. G., Nóbrega, A. M. C. D. S., & Amado, E. M. (2018). Capacity of tissue water regulation is impaired in an osmoconformer living in impacted estuaries? *Ecotoxicology and Environmental Safety*, 166, 375–382. <https://doi.org/10.1016/j.ecoenv.2018.09.111>
- Eerkes-Medrano, D., & Thompson, R. (2018). *Occurrence, fate, and effect of microplastics in freshwater systems* (E. Y. Zeng, Ed.; pp. 95–132). Elsevier. <https://doi.org/10.1016/B978-0-12-813747-5.00004-7>
- Evans, D. H. (1987). *The Fish Gill: Site of Action and Model for Toxic Effects of Environmental Pollutants*.
- Gouveia, R. S. D., Lira, G. L. D. A., Anselmo Ramos, T. P., & Medeiros, E. S. F. (2017). Ichthyofauna of the Reserva Biológica Guaribas and surrounding areas, state of Paraíba, Brazil. *Check List*, 13(5), 581–590. <https://doi.org/10.15560/13.5.581>
- Haredi, A. M. M., Mourad, M., Tanekhy, M., Wassif, E., & Abdel-Tawab, H. S. (2020). Lake edku pollutants induced biochemical and histopathological alterations in muscle tissues of Nile tilapia (*Oreochromis niloticus*). *Toxicology and Environmental Health Sciences*, 12(3), 247–255. <https://doi.org/10.1007/s13530-020-00042-w>
- Harmon, S. M. (2018). *The effects of microplastic pollution on aquatic organisms* (E. Y. Zeng, Ed.; pp. 249–270). Elsevier. <https://doi.org/10.1016/B978-0-12-813747-5.00008-4>
- Heinrichs, R., Otto, R., Magalhães, A., & Meirelles, G. C. (2017). *Importance of sugarcane in Brazilian and world bioeconomy* (S. Dabbert, I. Lewandowski, J. Weiss, & A. Pyka, Eds.; pp. 205–217). Springer International Publishing. https://doi.org/10.1007/978-3-319-58374-7_11
- Kurelec, B., Smital, T., Pivèviæ, B., Eufemia, N., & Epel, D. (2000). Multixenobiotic resistance, p-glycoprotein, and chemosensitizers. *Ecotoxicology*, 9(5), 307–327. <https://doi.org/10.1023/A:1026560922731>
- Lê, S., Josse, J., & Husson, F. (2008). FactoMineR: An R package for multivariate analysis. *Journal of Statistical Software*, 25(1), 1–18.
- Macêdo, A. K. S., Da Silva, J. R. P., Dos Santos, H. B., Thomé, R. G., Vendel, A. L., & Amado, E. M. (2019). Estuarine fish assemblages present a species-specific difference in the multixenobiotics resistance activity. *Journal of Experimental Zoology Part A: Ecological and Integrative Physiology*, 331(10), 530–539. <https://doi.org/10.1002/jez.2320>
- Mackereth, F. J. H., Heron, J., & Talling, J. F. (1978). Water analysis: Some revised methods for limnologists. (scientific publication n. 36) freshwater biological association. *Cumbria and Dorset, England*.

- Maitland, P. S. (1990). *Field Studies: Sampling in Freshwaters* (pp. 123–148). Springer US. https://doi.org/10.1007/978-1-4613-0501-9_5
- Marques, M. C. M., & Grelle, C. E. V. (Eds.). (2021). *The Atlantic Forest: History, Biodiversity, Threats and Opportunities of the Mega-diverse Forest*. Springer International Publishing. <https://doi.org/10.1007/978-3-030-55322-7>
- Marques, M. C. M., Trindade, W., Bohn, A., & Grelle, C. E. V. (2021). *The Atlantic Forest: An Introduction to the Megadiverse Forest of South America* (M. C. M. Marques & C. E. V. Grelle, Eds.; pp. 3–23). Springer International Publishing. https://doi.org/10.1007/978-3-030-55322-7_1
- Masullo, Y. A. G., Gurgel, H. D. C., Laques, A.-E., & Carvalho, D. S. (2020). Avaliação da efetividade em unidades de conservação: um estudo de caso no estado do maranhão, Brasil. *Revista Tamoios*, 16(3). <https://doi.org/10.12957/tamoios.2020.51272>
- McCune, B., & Grace, J. B. (2002). *Analysis of ecological communities*. MjM Software Design.
- Medeiros, E. S. F., Silva, M. J., & Ramos, R. T. C. (2008). Application of catchment- and local-scale variables for aquatic habitat characterization and assessment in the brazilian semi-arid region. *Neotropical Biology and Conservation*, 3(1), 13–20.
- Medeiros, E. S. F., Silva, M., Figueiredo, B., Ramos, T., & Ramos, R. (2010). Effects of fishing technique on assessing species composition in aquatic systems in semi-arid brazil. *Brazilian Journal of Biology*, 70, 255–262. <https://doi.org/10.1590/S1519-69842010000200004>
- Miranda, J. (2012). Ameaças aos peixes de riachos da mata atlântica. *Ameaças Aos Peixes de Riachos Da Mata Atlântica*.
- MMA. (2003). *Plano de manejo reserva biológica guaribas. Fase 2*. <https://bit.ly/4gtUh3i>
- MMA. (2026). *Painel de unidades de conservação brasileiras*. <https://encurtador.com.br/kLLe>
- Mugodo, J., Kennard, M. J., Liston, P., Nichols, S., Linke, S., Norris, R. H., & Lintermans, M. (2006). Local stream habitat variables predicted from catchment scale characteristics are useful for predicting fish distribution. *Hydrobiologia*, 572(1), 59–70. <https://doi.org/10.1007/s10750-006-0252-7>
- Myers, N., Mittermeier, R. A., Mittermeier, C. G., Da Fonseca, G. A. B., & Kent, J. (2000). Biodiversity hotspots for conservation priorities. *Nature*, 403(6772), 853–858. <https://doi.org/10.1038/35002501>
- Ondei, L. D. S., Teresa, F. B., Garcia, D. P., Felício, A. A., Da Silva, D. G. H., & De Almeida, E. A. (2020). Fish biomarker responses to perturbation by drought in streams. *Neotropical Ichthyology*, 18(2), e190130. <https://doi.org/10.1590/1982-0224-2019-0130>
- Padial, A. A., Costa, A. P. L., Bonecker, C. C., Nogueira, D. G., Roque, F. O., Message, H. J., Diniz, L. P., Cavalcante, L. L., Vallejos, M. A. V., Ota, R. R., & Ludwig, T. A. V. (2021). *Freshwater Studies in the Atlantic Forest: General Overview and Prospects* (M. C. M. Marques & C. E. V. Grelle, Eds.; pp. 205–230). Springer International Publishing. https://doi.org/10.1007/978-3-030-55322-7_10
- Peel, M. C., Finlayson, B. L., & McMahon, T. A. (2007). Updated world map of the Köppen-Geiger climate classification. *Hydrol. Earth Syst. Sci.*
- Pereira, K. M. G., Cordeiro, N. G., Suzuki, L. P. Z. L., & Berg, E. van den. (2026). How important are small fragments for restoration in the central portion of the atlantic forest? *Biological Conservation*, 321, Article 111958. <https://doi.org/https://doi.org/10.1016/j.biolcon.2026.111958>

biocon.2026.111958

- Piedras, S., Oliveira, J., Moraes, P., & Bager, A. (2006). Toxicidade aguda da amônia não ionizada e do nitrito em alevinos de cichlasoma facetum (jenyns, 1842). *Ciência e Agrotecnologia*, 30, 1008–1012. <https://doi.org/10.1590/S1413-70542006000500027>
- Pusey, B., Kennard, M. J., & Arthington, A. (2004). *Study area, data collection, analysis and presentation* (B. Pusey, M. J. Kennard, & A. Arthington, Eds.; pp. 26–48). CSIRO Publishing.
- R Core Team. (2017). *R: A language and environment for statistical computing*. <https://www.R-project.org/>
- R Studio Team. (2022). *RStudio: Integrated development environment for r*. RStudio, PBC. <https://posit.co/products/open-source/rstudio/>
- Sabae, S. Z., & Mohamed, F. A. S. (2015). *Effect of Environmental Pollution on the Health of Tilapia spp. from Lake Qarun*.
- Santos, M. B., Monteiro Neto, I. E., Souza Melo, S. R. C., & Amado, E. M. (2017). Hemolymph and gill carbonic anhydrase are more sensitive to aquatic contamination than mantle carbonic anhydrase in the mangrove oyster *Crassostrea rhizophorae*. *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*, 201, 19–25. <https://doi.org/10.1016/j.cbpc.2017.08.008>
- Smital, T., & Kurelec, B. (1998). The chemosensitizers of multixenobiotic resistance mechanism in aquatic invertebrates: a new class of pollutants. *Mutation Research - Fundamental and Molecular Mechanisms of Mutagenesis*, 399(1), 43–53. [https://doi.org/10.1016/S0027-5107\(97\)00265-0](https://doi.org/10.1016/S0027-5107(97)00265-0)
- Smital, T., Luckenbach, T., Sauerborn Klobucar, R., Hamdoun, A., Vega, R., & Epel, D. (2004). Emerging contaminants - pesticides, PPCPs, microbial degradation products and natural substances as inhibitors of multixenobiotic defense in aquatic organisms. *Mutation Research*, 552, 101–117. <https://doi.org/10.1016/j.mrfmmm.2004.06.006>
- Sokal, R. R., & Rohlf, F. J. (1995). *Biometry: The principles and practice of statistics in biological research* (3rd ed.). W.H. Freeman; Company.
- Soler, J. M. P. (2004). *Planejamento de experimentos e pesquisa em limnologia* (pp. 16–24). BICUDO, C. E. M. BICUDO, D. C.
- Žaja, R., Klobučar, G. I. V., Sauerborn Klobučar, R., Hackenberger, B. K., & Smital, T. (2006). Haemolymph as compartment for efficient and non-destructive determination of p-glycoprotein (pgp) mediated MXR activity in bivalves. *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*, 143(1), 103–112. <https://doi.org/10.1016/j.cbpc.2005.12.009>
- Zar, J. H. (2010). *Biostatistical Analysis* (5th ed.). Prentice Hall.

Figures and Tables

Figure 1: Location of sampling sites along the Barro Branco Stream (Mamanguape, PB) (Site 1 06°43'06"S and 35°10'54"W; Site 2 06°42'38"S and 35°10'38"W; Site 3 06°43'06"S and 35°10'54"W; Site 4 06°41'46"S and 35°10'36"W; Site 5 06°40'56"S and 35°10'27"; Site 6 06°40'18"S and 35°10'34"W). Site 1 is the reference site for this study.

Figure 2: Images of the main four types of sites found along the Barro Branco stream (Mamanguape, PB). Type I, well preserved and within the conservation unit. Type II site, preserved and outside the conservation unit. Type III site, disturbed but withing the conservation unit. Type IV site, disturbed and outside the conservation unit.

Figure 3: Principal Component Analysis of the environmental variables for the Barro Branco stream (Mamanguape, PB).

Figure 4: Counts of number of individuals per species and biomarker use (MXR, Multixenobiotic resistance phenotype activity; Moisture, Plasmatic protein dosage and moisture content) for each study site.

Figure 5: Total plasmatic proteins (mg) variance by sampling site and fish species.

Figure 6: Gill and muscle moisture content (%) by sampling site and fish species.

Figure 7: Average fluorescence of rhodamin B expressed by relative fluorescence units/milligrams of gill tissue of fish by sampling site.

Table 1: Environmental variables averaged for the sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB). Site 1 is the reference site for this study.

Figures

Fig.: Location of sampling sites

Fig.: Images of the main four types of sites

Fig.: Principal Component Analysis of the environmental variables

Fig.: Counts of number of individuals per species and biomarker

Fig.: Total protein variance

Fig.: Moisture content of tissues

Fig.: Average fluorescence for rhodamin B

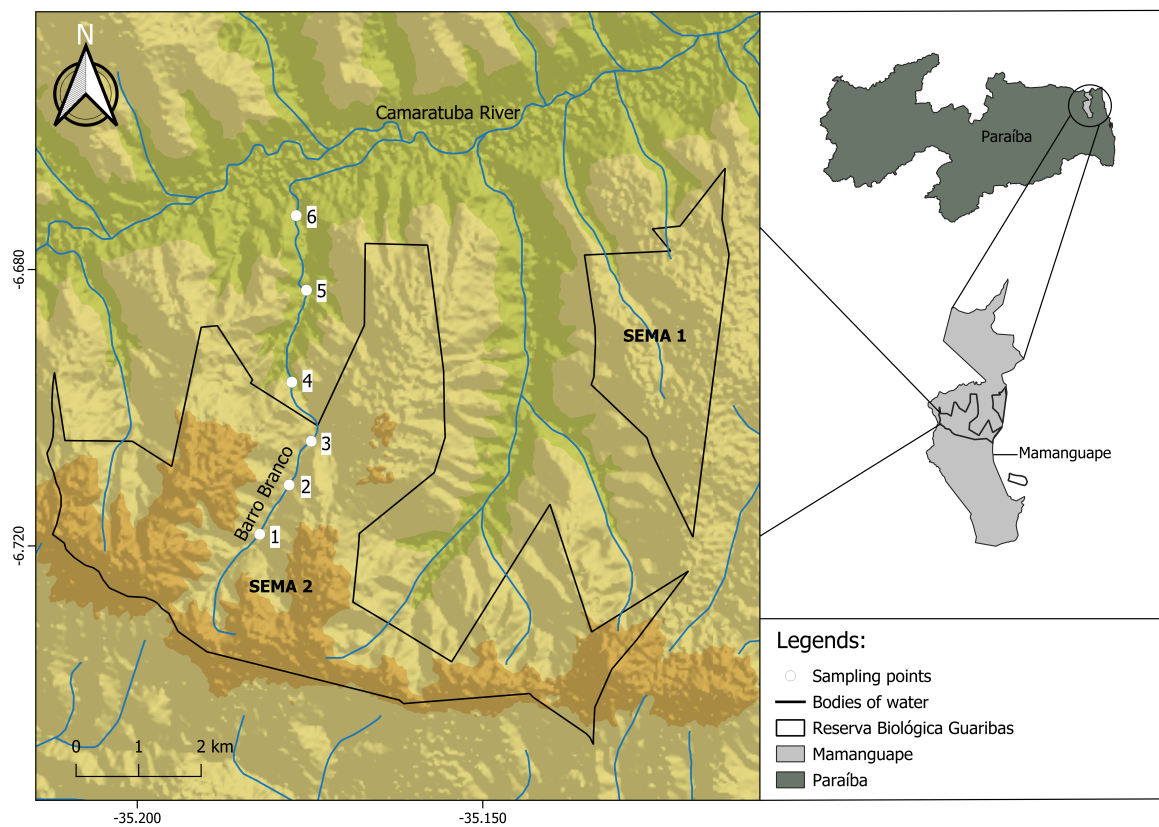


Figure 1: Location of sampling sites.

Type I



Type II



Type III



Type IV



Figure 2: Images of the main four types of sites.

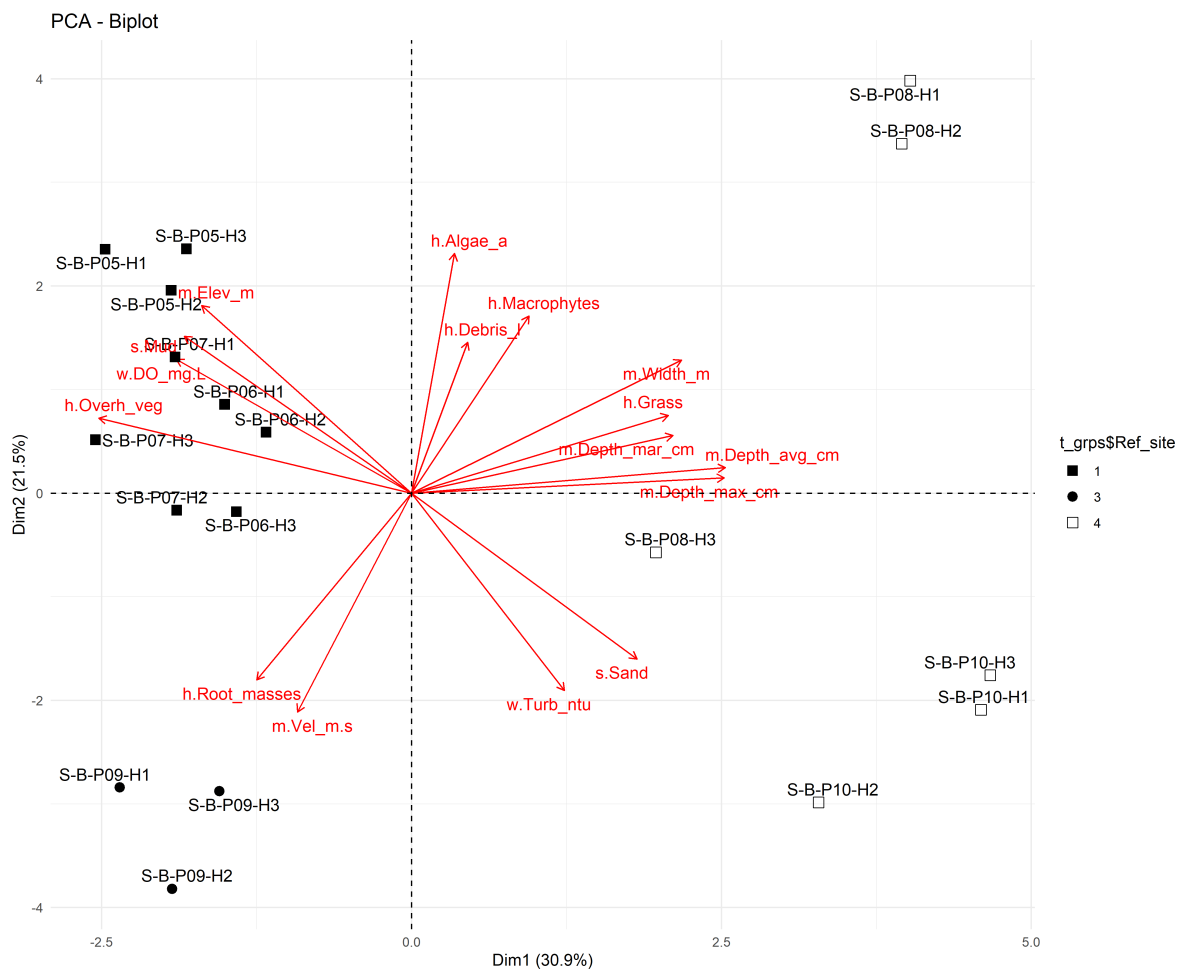


Figure 3: Principal Component Analysis of the environmental variables.

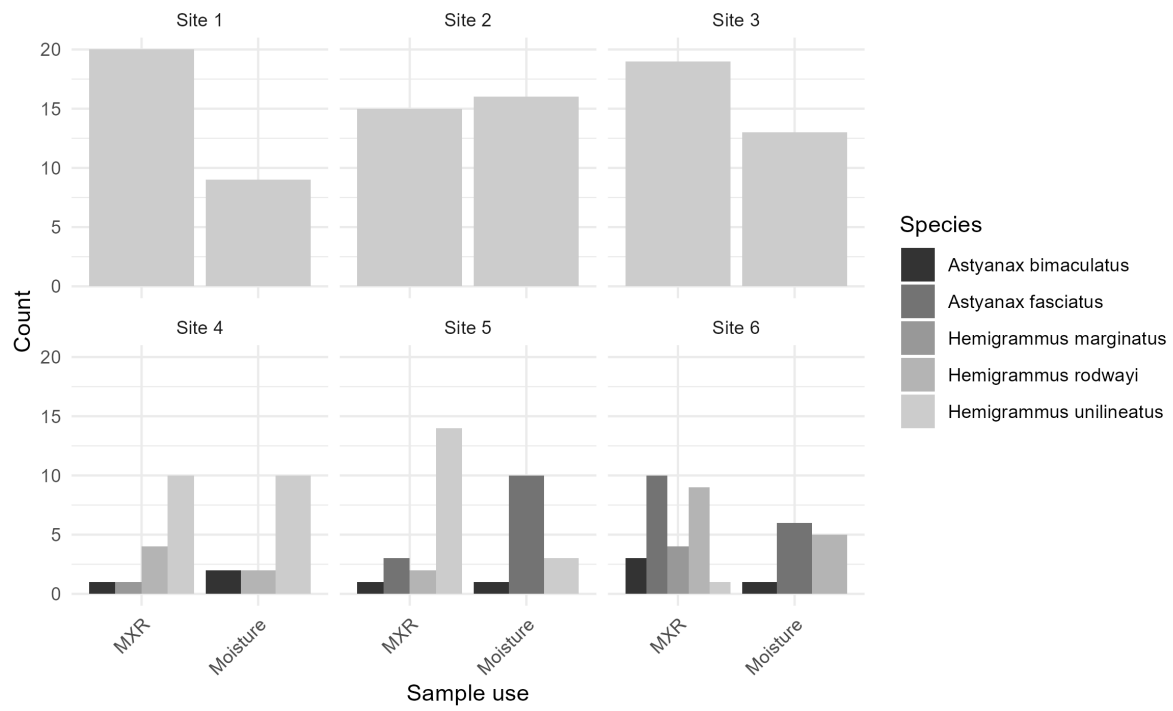


Figure 4: Counts of number of individuals per species and biomarker.

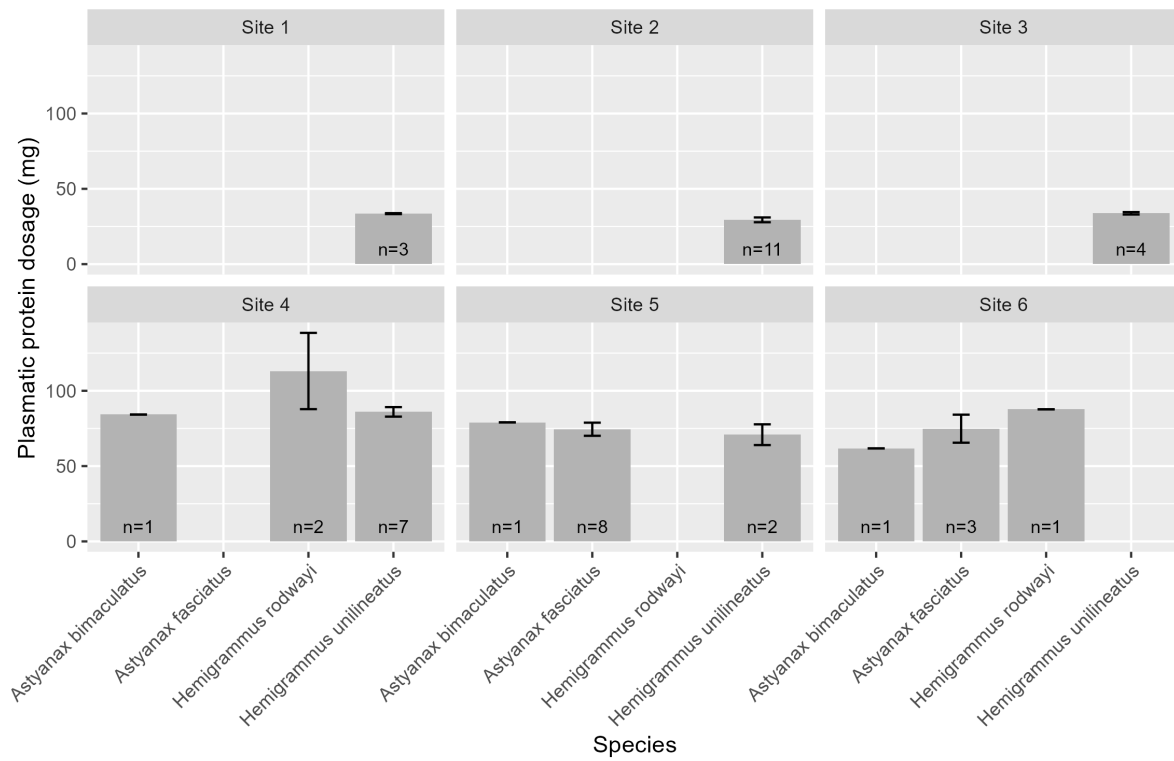


Figure 5: Total plasmatic proteins variance.

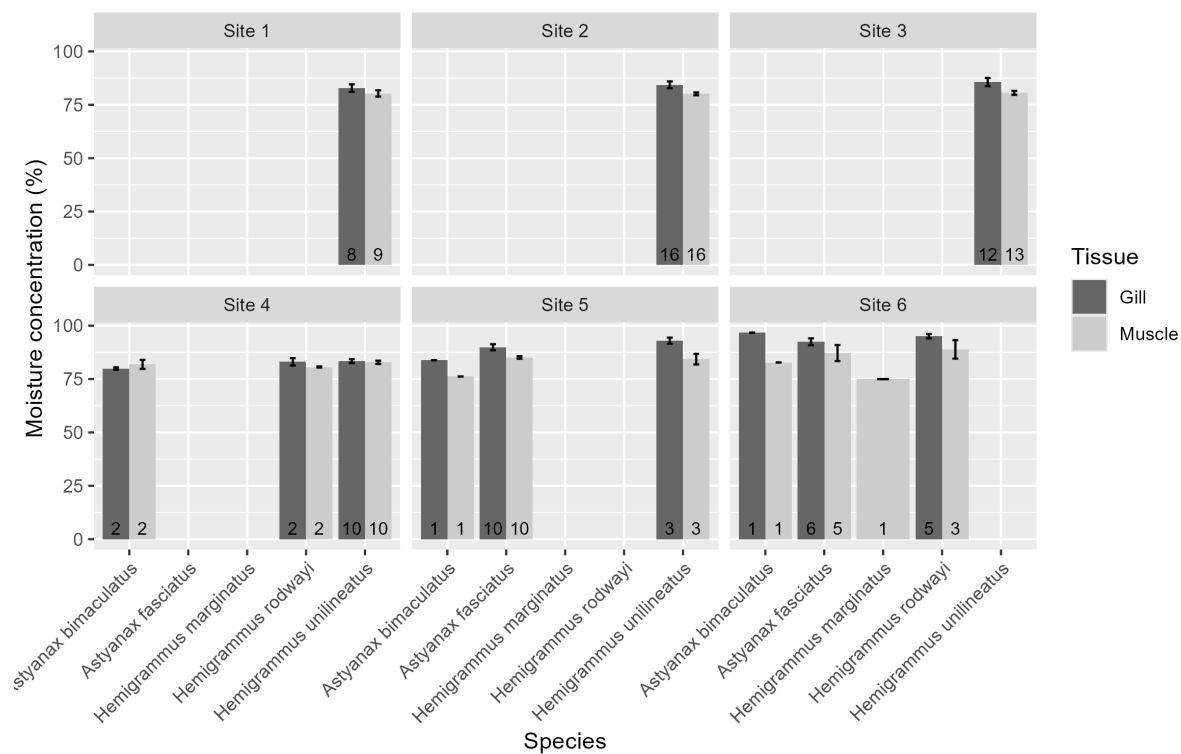


Figure 6: Gill and muscle moisture content.

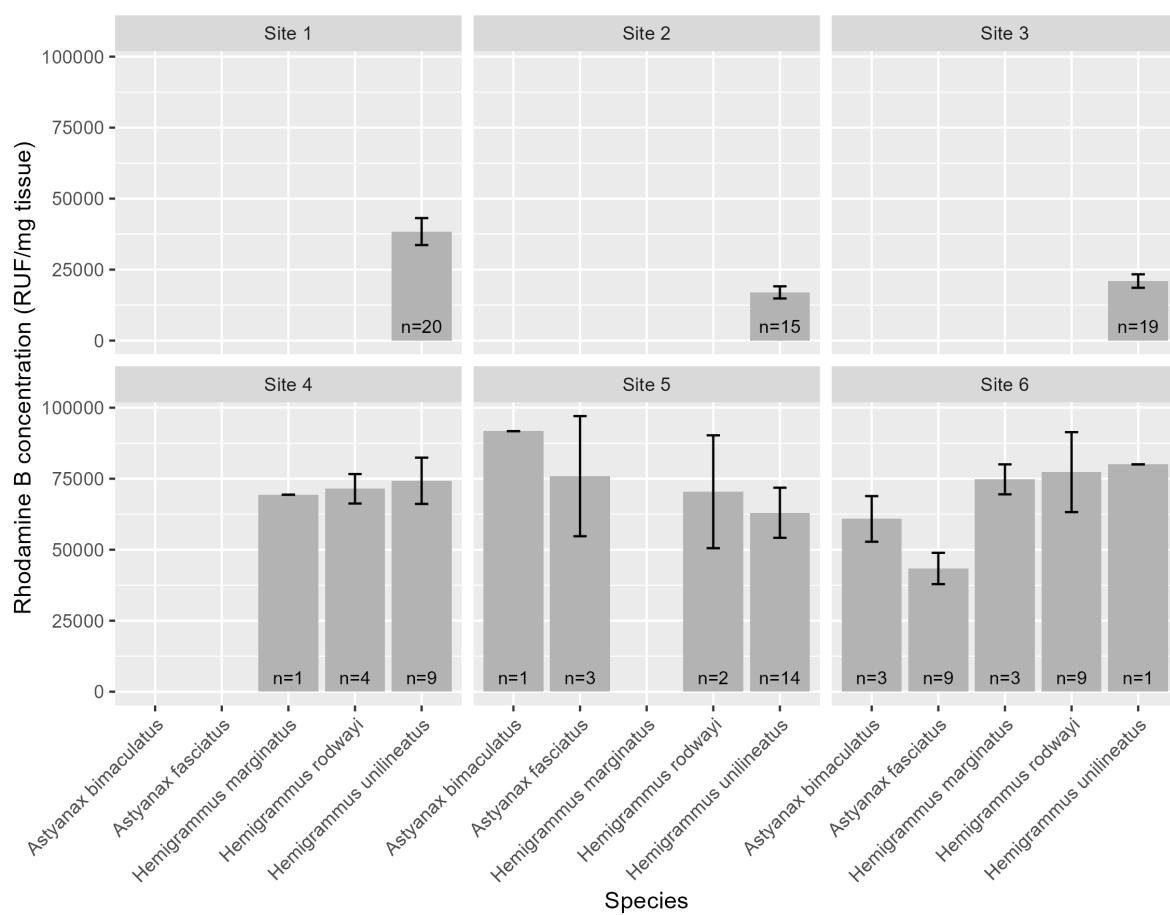


Figure 7: Average fluorescence of rhodamin B.

Table 1: Environment data GT table.

Variable	Site 1	Site 2	Site 3	
Water quality				
Temperature (°C)	25.8 (24.9-27.3)	26.5 (25.3-27.5)	25.2 (24.9-25.5)	27.8
Dissolved oxygen (mg/L)	8.1 (7.8-8.5)	7.5 (7.2-7.8)	7.9 (7.7-8)	7.5
Conductivity (uS/cm)	146.4 (134-164.4)	148.3 (134.1-156.8)	129 (98.3-160.2)	160.6 (158)
Turbidity (NTU)	7.4 (4.6-10.7)	6.1 (2.1-14.1)	4.4 (1.5-9.2)	6.4
pH	7 (6.3-7.5)	6.8 (5.9-7.7)	5.5 (4.6-6)	5.
Morphology				
Elevation (m a.s.l.)	107.1 (107.1-107.1)	112.8 (112.8-112.8)	89 (89-89)	89.1 (8)
Width (m)	3.6 (3.6-3.6)	3.7 (2.3-4.4)	1.7 (1.1-2.1)	14.6 (1)
Water velocity (m/s)	0.017 (0-0.025)	0.035 (0.035-0.035)	0.073 (0.022-0.108)	
Slope	1.2 (1-1.5)	2.5 (2.5-2.5)	2.2 (1.5-2.5)	
Average depth (cm)	13.7 (13.7-13.7)	23 (20.3-28.3)	19.7 (16-22.7)	44.2
Substrate composition (%)				
Mud	100 (100-100)	100 (100-100)	90 (90-90)	80
Sand	0 (0-0)	0 (0-0)	10 (10-10)	20
Habitat structures (%)				
Aquatic macrophytes	5 (5-5)	20 (0-30)	0 (0-0)	35.8
Littoral grass	0 (0-0)	5 (5-5)	2 (0-5)	39.2
Submerged vegetation	5 (5-5)	8.3 (5-15)	3.3 (0-10)	
Overhanging vegetation	90 (90-90)	81.7 (80-85)	85 (80-90)	30
Leaf litter	10 (10-10)	3.7 (1-5)	20 (5-30)	26.7
Algae	5 (5-5)	0.3 (0-1)	11 (2-21)	16.7
Debris	30 (30-30)	5.3 (5-6)	10.3 (5-15)	31.7
Root masses	0 (0-0)	8.3 (5-15)	10 (5-15)	

Tables

Tab.: Environmental data table

Apendices

Non-used figures and tables

Non-used codes

```
1 table(1:10)
1 library(readr)
2 library(dplyr)
3 library(gt)
4
5 m_wide_hab <- read.table("m_wide_hab.txt",
6                         sep = ";", dec = ".",
7                         row.names = 1,
8                         header = TRUE,
9                         na.strings = NA)
10
11 #m_mapa <- read_tsv("column_labels.txt", show_col_types = FALSE)
12 dados <- m_wide_hab
13 dados
14 nomes <- names(dados)[-1]
15 nomes
16
17 df_nomes <- data.frame(
18   original = nomes,
19   final    = nomes,
20   stringsAsFactors = FALSE)
21
22 df_nomes
23 df_nomes$final <- gsub("_", "<br>", df_nomes$final)
24 dados <- mutate(dados, across(-Variable, ~ gsub(" \\(", "<br>(", .)))
25 #fix(df_nomes)
26 #write.table(df_nomes, "df_nomes.txt")
27 df_nomes <- read.table("df_nomes.txt")
28
29 labels_finais <- setNames(
30   lapply(df_nomes$final, md),
31   nomes)
32
33 tabela_gt <- dados %>%
```

```

34   gt(rowname_col = "Variable") %>%
35   #   tab_header(
36   #     title = "Características Ambientais",
37   #     subtitle = "Valores apresentados como média (mín-máx)"
38   #   ) %>%
39   fmt_markdown(
40     columns = -Variable) %>%
41   cols_align(
42     align = "right",
43     columns = -Variable) %>%
44   cols_label(.list = labels_finais)
45
46   tabela_gt
47   gtsave(tabela_gt, "gt-hab_gtable.html")
48   gtsave(tabela_gt, "fig-hab_gtable.png")
49   saveRDS(tabela_gt, "gt-hab_gtable.rds")

```